

Coverage optimization strategy for 3D wireless sensor network based on adaptive inertia weight Arctic Puffin Optimization Algorithm

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Abstract

A multi-strategy improved Arctic Puffin Optimization Algorithm (CAAPO) based node coverage optimization approach is provided to tackle the problems of unequal node deployment and low coverage in wireless sensor networks (WSNs). Drawing inspiration from the species' survival strategies in polar environments, the methodology incorporates three key enhancements: Tent chaotic mapping for initial solution refinement, lens imaging inversion mechanisms to prevent premature convergence through population diversification, and adaptive inertia weighting for balanced exploration-exploitation dynamics across optimization phases. Empirical validation through CEC2017 benchmarks and engineering simulations demonstrates CAAPO's efficacy in nodal deployment optimization. Implementation within sensor coverage models achieves maximal coverage through strategic node positioning, exhibiting superior spatial distribution uniformity compared to conventional methods. The algorithm demonstrates operational efficiency through minimized computational overhead and energy expenditure, thereby enhancing network longevity in resource-constrained environments.

Keywords: wireless sensor networks, Arctic Puffin Optimization, node deployment, chaos mapping, adaptive inertia weights, coverage rate

1. Introduction

The widespread adoption of 5G-powered IoT infrastructure has accelerated the large-scale implementation of wireless sensor networks (WSNs) across industrial sectors, with notable applications in maritime communication systems and power grid monitoring frameworks (Temene et al., 2022; Gao et al., 2023; Rani et al., 2023). A WSN is defined as an interconnected system of autonomous sensor nodes equipped with embedded functionalities for distributed computing, data storage, and wireless communication. Experimental research indicates that the spatial coverage efficiency of nodes is a key determinant of overall network performance (El-shafeiy et al., 2021; Rahaman & Azharuddin, 2022). Strategic optimization of sensor node architecture has become a crucial approach to addressing two core challenges: (1) alleviating resource redundancy stemming from random deployment patterns, and (2) lowering capital costs associated with network infrastructure development (Ahmad et al., 2024; Jaafari et al., 2022). By optimizing geometric configurations, improvements in node distribution uniformity and expansion of coverage ranges can be achieved, thereby enhancing spatial resource utilization efficiency while boosting the accuracy of environmental perception and the reliability of data acquisition (Gad, 2022; Anka, 2025). This dual-benefit strategy concurrently strengthens network resilience and operational sustainability and reduces the total cost of ownership (Kivi & Majidnezhad, 2022; Cao et al., 2023).

The problem of coverage optimization represents a critical research frontier in WSN deployments, directly influencing network operational effectiveness and service quality (Bai et al., 2023). Existing studies classify WSN coverage challenges using multiple taxonomies based on distinct operational criteria; spatial dimensionality [including one-dimensional (1D) linear, two-dimensional (2D) planar, and three-dimensional (3D) volumetric configurations] and coverage quality requirements (such as minimum threshold coverage, maximum area coverage, and K-coverage redundancy models) (Cao et al., 2023; Tossa et al., 2022; Yue et al., 2024). Contemporary solution methods can be divided into two primary categories: geometric modelling approaches that utilize topological representations for coverage analysis, and computational optimization algorithms that employ metaheuristic techniques to approximate Pareto-optimal solutions. The latter demonstrate greater applicability in addressing the inherent operational constraints of WSNs, including large-scale node distribution, terrain heterogeneity, and dynamic environmental conditions (Kiani & Seyyedabbasi, 2022; Jia et al., 2023; Yue et al., 2023). Traditional gradient-based optimization methods exhibit three fundamental limitations in this context: vulnerability to local optima, computational infeasibility, and inadequacy for real-time processing (Chen et al., 2024).

Metaheuristic algorithms, particularly genetic algorithms (GA), particle swarm optimization (PSO), and simulated annealing (SA), overcome these limitations through three evolutionary

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mechanisms: (1) global search capability via biological evolution emulation (GA), swarm intelligence simulation (PSO), or thermodynamic process modelling (SA); (2) model-independent stochastic solution space exploration; and (3) adaptive operator implementation for exploration-exploitation balance (Yue et al., 2023; Jiang et al., 2024). GA's evolutionary operators systematically refine solution quality through selection pressure, crossover recombination, and mutation probability mechanisms, effectively circumventing convergence stagnation (Hanh et al., 2019; Bahadur & Lakshmanan, 2023; Yarinezhad & Hashemi, 2023). PSO algorithm achieves rapid global optimum approximation through particle velocity modulation and neighbourhood best tracking (Du et al., 2019; Zhang, 2020; Chen & Liang, 2022; Ding et al., 2023; Siamantas & Kandris, 2024). These methodologies enable efficient multi-objective sensor deployment optimization in dynamic coverage scenarios, simultaneously ensuring monitoring comprehensiveness, energy efficiency maximization, and resource utilization optimization.

Contemporary research indicates that WSN coverage optimization in 1D and 2D environments has reached operational maturity, with mainstream algorithms achieving near-saturation coverage efficiency (Ammari, 2021; Yang et al., 2022). However, 3D spatial coverage remains a persistent challenge in practical engineering applications, characterized by unresolved issues in convergence dynamics, computational complexity, and nodal redundancy (Kiani, 2025; Priyadarshi & Gupta, 2021; Walia et al., 2022). Current methodologies for 3D WSN deployment exhibit three critical limitations: (1) suboptimal convergence rates in evolutionary algorithms, (2) polynomial-time complexity scaling with nodal density, and (3) inefficient resource utilization patterns. Representative approaches include the wavelet transform-based deployment strategy (WTDS) (Zhang et al., 2021), which integrates GA mutation operators with terrain-adaptive sensor relocation, demonstrating enhanced coverage quality through empirical validation against Delaunay triangulation baselines (Du, 2020). Temel's hybrid methodology synergises cat swarm optimization (CSO) with line-of-sight (LOS) perception modelling, achieving superior deployment accuracy in obstructed 3D environments compared to the GA algorithm (Temel et al., 2013; Li et al., 2021). Notwithstanding these advancements, contemporary deployment paradigms predominantly emphasize coverage maximization while neglecting network longevity considerations. Cao et al.'s distributed particle swarm optimization (DPSO) algorithm addresses this gap through multi-objective optimization in obstructed 3D environments, incorporating reliability-constrained lifetime modelling to simultaneously maximize coverage and operational sustainability (Gou et al., 2023).

In summary, while significant advancements have been made in WSNs coverage optimization methodologies, persistent limitations constrain practical implementation efficacy (Nematzadeh et al., 2023). Current approaches predominantly employ Boolean sensing models that rely on oversimplified environmental interaction abstractions, resulting in limited applicability in real-world deployment scenarios (Li et al., 2022). Furthermore, the inherent complexity of 3D operational environments necessitates fundamental algorithmic improvements in coverage optimization mechanisms (Zheng et al., 2023; Li et al., 2024). This investigation proposes an Enhanced Arctic Puffin Optimization Algorithm (CAAPO) for sensor node distribution optimization, inspired by the species' evolutionary survival strategies in polar ecosystems (Yao et al., 2023). The metaheuristic algorithm mathematically models three behavioural patterns observed in Arctic puffin colonies: collective foraging dynamics during migratory navigation, social

coordination principles in breeding activities, and adaptive environmental interaction mechanisms (Singh et al., 2022).

Although the APO algorithm has good convergence speed and accuracy, it has a common problem that as the algorithm runs to the later stages, the population diversity gradually decreases, and once it falls into a local optimum, it is difficult to escape. In response to the problem of difficulty in escaping local optima in the later stage of the APO algorithm, this paper adopts a lens imaging reverse learning strategy to expand the search range and improve the algorithm's ability to escape local optima. In the development phase, an adaptive inertia factor is introduced to balance the ability of global exploration and local optimization, ensuring that the algorithm can effectively utilize the current best solution and guide the search direction. In order to improve the convergence speed of the algorithm and enhance its optimization ability, this paper introduces Tent chaotic mapping during initialization to improve the quality of the initial solution and enhance the overall performance of the algorithm. Finally, the improved algorithm was compared and analysed with the original algorithm and six classical algorithms through simulation experiments on the CEC2017 test set. Then, two engineering application comparison experiments and 2D and 3D sensor deployment analysis experiments were conducted to verify the effectiveness of the proposed multi-strategy improved algorithm in sensor node deployment.

2 Mathematical Model of Coverage for Wireless Sensor Networks

2.1. Perception model

In order to simplify the complexity of the perception model, such challenges are frequently reduced to binary models, also known as Boolean perception models (Li & Liu, 2020). In a 2D plane, a sensor node's sensing range is a circular area produced by centring the sensor node and setting R as the radius of the circle. The likelihood of being felt inside this range is one, but the region outside of this range has a probability of zero. The mathematical model goes as follows:

$$P(r, s_i) = \begin{cases} 1, & \text{if } d(r, s_i) \leq R \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

In the formula, $P(r, s_i)$ represents the probability of the sensor detecting the point and $d(r, s_i)$ represents the Euclidean distance from the sensor to the point.

2.2. Coverage

Current research predominantly employs area grid models to analyse WSN coverage, with regional coverage rate serving as the standard metric for evaluating optimization efficacy (Figure 1; Wang et al., 2017). This parameter quantifies the proportion of the monitored area effectively serviced by sensor nodes.

Now, assume there are n sensors in the sensor network, and their set is represented as

$$S = \{s_i\}, i = 1, \dots, n. \quad (2)$$

In the formula, S represents the sensor network, s_i represents the sensor nodes, and n is the number of sensor nodes in the network.

The area coverage rate is defined as the ratio of the effective coverage area of all sensor nodes to the total area of the target

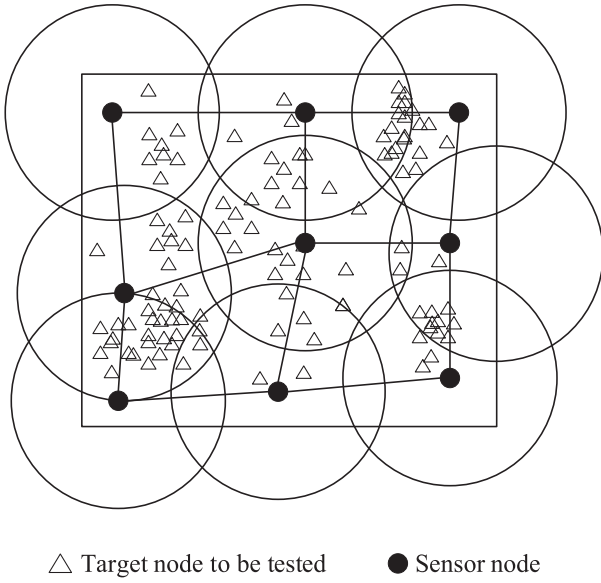


Figure 1: Coverage structure of WSNs.

region, as shown in the following formula:

$$\eta = \frac{\sum_{i=1, \dots, n} \Phi_{s_i}}{A}. \quad (3)$$

In the formula, η is the area coverage of the sensor network, Φ_{s_i} is the sensing area of the i th sensor, and A is the total area of the target region.

A specific point r in the target area may be perceived by different nodes in the network, so the probability of point r being perceived by the sensor network is

$$P(r, S) = 1 - \prod_{i=1, \dots, n} [1 - P(r, s_i)]. \quad (4)$$

Assuming there are m points in the area, the total coverage of the sensor network over all the sensed points in the area can be obtained as follows:

$$C(s) = \frac{\sum_{i=1, \dots, m} P(r, S)}{A}. \quad (5)$$

In the formula, $C(s)$ is the total coverage of all the sensed points in the target area by the sensor network.

The goal of this study is to deploy a given number of sensor nodes in the target region and, using various swarm intelligence methods to optimize the layout of the nodes, increase the WSN's coverage to the maximum possible value.

3. Standard Arctic Puffin Optimization Algorithm

The Arctic Puffin Optimization (APO) algorithm is a novel meta-heuristic approach inspired by the species' bimodal foraging behaviour (Wang et al., 2024). This bio-inspired framework emulates two key foraging strategies: aerial search for global exploration of solution space and aquatic predation for local refinement of candidate solutions. Through this dual-phase mechanism, APO demonstrates enhanced capability in addressing complex optimization problems (Fakhouri et al., 2024).

3.1. Aerial search phase

(1) Aerial search

The Arctic puffin primarily lives in groups, and when they go out to hunt, they often fly in formation to search for prey. This cooperative hunting method dramatically increases the chances of successful predation.

The position update formula is as follows:

$$\vec{Y}_i^{t+1} = \vec{X}_i^t + (\vec{X}_i^t - \vec{X}_r^t) * L(D) + R, \quad (6)$$

$$R = \text{round}(0.5 * (0.05 + \text{rand})) * a, \quad (7)$$

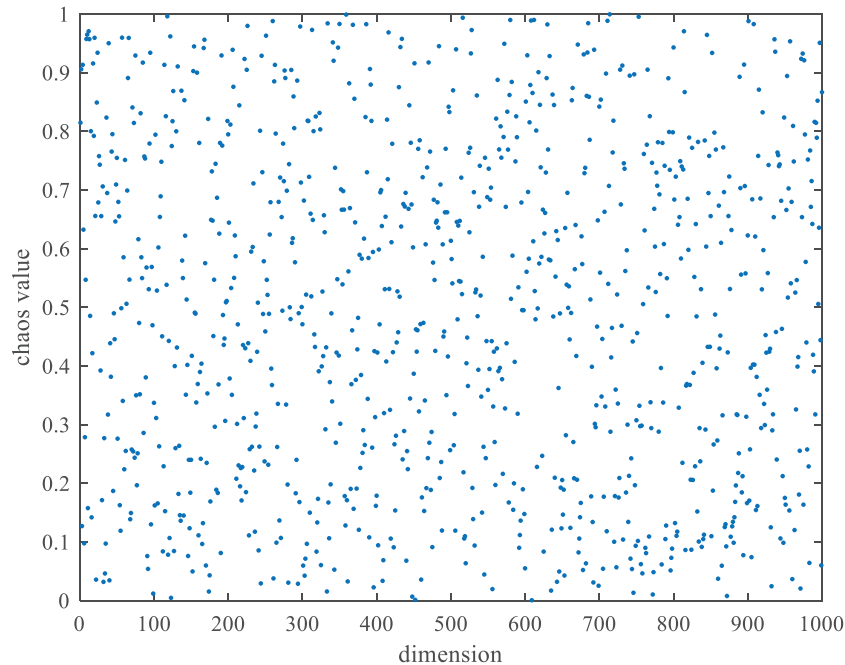


Figure 2: Tent chaos mapping sequence.

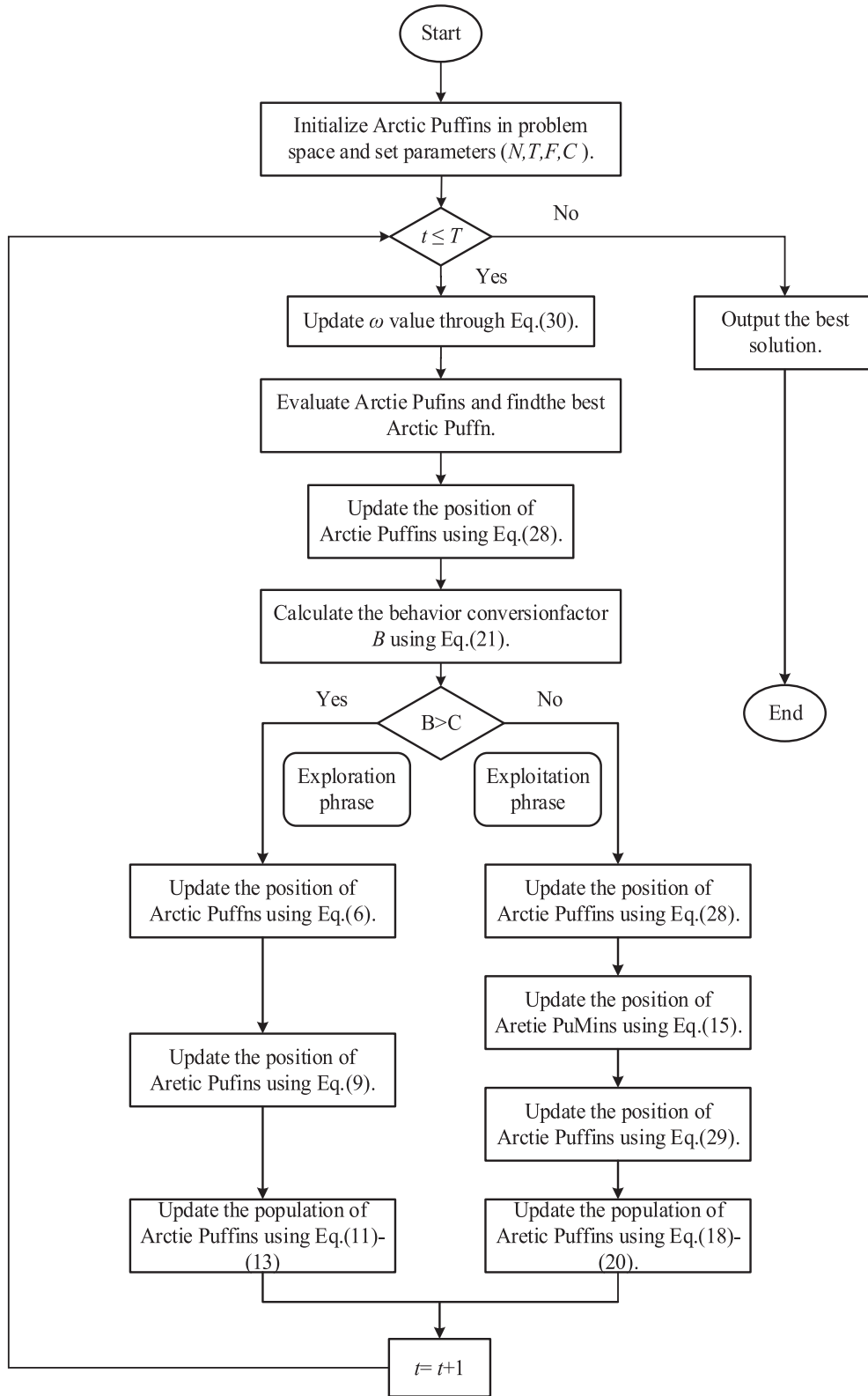


Figure 3: The flowchart and pseudocode of the CAAPO algorithm.

$$\alpha \sim \text{Normal}(0, 1). \quad (8)$$

In the formula: r is a random number between $[1, N - 1]$, $\vec{X}_i^t$ represents the i th individual in the current population, and $\vec{X}_r^t$ represents an individual randomly selected from the current popula-

tion. $L(D)$ is the random number generated by Levy flight, D is the dimension constant; α is the random number that follows a normal distribution.

The Levy flight utilized in the method is primarily used to imitate the Arctic puffin's flying process, with the Levy flight

Table 1: CEC2017 Partial Test Functions.

	No.	Functions	$F_i = F_i(x)$
Unimodal functions	1	Shifted and rotated bent cigar function	100
	2	Shifted and rotated sum of different power function*	200
	3	Shifted and rotated Zakharov function	300
Simple multimodal functions	4	Shifted and rotated Rosenbrock's function	400
	5	Shifted and rotated Rastrigin's function	500
	6	Shifted and rotated expanded scaffer's F6 function	600
	7	Shifted and rotated Lunacek Bi_Rastrigin function	700
	8	Shifted and rotated non-continuous Rastrigin's function	800
	9	Shifted and rotated Levy function	900
	10	Shifted and rotated Schwefel's function	1000
	11	Hybrid function 1 ($N = 3$)	1100
Hybrid functions	12	Hybrid function 2 ($N = 3$)	1200
	13	Hybrid function 3 ($N = 3$)	1300
	14	Hybrid function 4 ($N = 4$)	1400
	15	Hybrid function 5 ($N = 4$)	1500
	16	Hybrid function 6 ($N = 4$)	1600
	17	Hybrid function 6 ($N = 5$)	1700
	18	Hybrid function 6 ($N = 5$)	1800
	19	Hybrid function 6 ($N = 5$)	1900
	20	Hybrid function 6 ($N = 6$)	2000
	21	Composition function 1 ($N = 3$)	2100
Composition functions	22	Composition function 2 ($N = 3$)	2200
	23	Composition function 3 ($N = 4$)	2300
	24	Composition function 4 ($N = 4$)	2400
	25	Composition function 5 ($N = 5$)	2500
	26	Composition function 6 ($N = 5$)	2600
	27	Composition function 7 ($N = 6$)	2700
	28	Composition function 8 ($N = 6$)	2800
	29	Composition function 9 ($N = 3$)	2900
	30	Composition function 10 ($N = 3$)	3000

Search range: $[-100, 100]^D$

coefficient adjusting the Arctic puffin's locations over time. This method can significantly improve their prey-finding efficiency. Furthermore, when Arctic puffins confront airborne predators such as seagulls, they circle to evade attacks.

(1) Pounce and prey

After the Arctic puffin locks onto its prey while flying in the air, it quickly alters its flight pattern and accelerates its wingbeats to dive towards the target prey. This rapid attack strategy can increase their chances of successful predation. At this stage, a speed coefficient S is introduced to adjust the state of the Arctic puffin during the diving process. The position update formula is as follows:

$$\vec{Z}_i^{t+1} = \vec{Y}_i^{t+1} * S, \quad (9)$$

$$S = \tan((rand - 0.5) * \pi). \quad (10)$$

During the airborne search phase, the Arctic puffin uses the Levy flight method to conduct a global search. On the other hand, by inserting the speed coefficient S , it balances the Arctic puffin's condition when diving for prey, resulting in the ideal individual and population during this phase. To better calculate and filter the ideal solution, the position update formulae for the two scenarios are integrated and changed as follows:

$$\vec{P}_i^{t+1} = \vec{Y}_i^{t+1} \cup \vec{Z}_i^{t+1}, \quad (11)$$

$$new = \text{sort}(\vec{P}_i^{t+1}), \quad (12)$$

$$\vec{X}_i^{t+1} = new(1 : N). \quad (13)$$

In the formula, sort means to sort by fitness values from smallest to largest and select the better population.

3.2. Underwater foraging phase

(1) Foraging

While aiding in hunting, the Arctic puffin frequently soars among schools of fish. This formation flying technique also makes it easier for the group to seek prey. When there is no target, the Arctic puffin will remain on the ground to observe other members' activities, detecting undersea heat sources and prey movements.

The position update formula is as follows:

$$\vec{W}_i^{t+1} = \begin{cases} \vec{X}_{r_1}^t + F * L(D) + (\vec{X}_{r_2}^t - \vec{X}_{r_3}^t), & rand \geq 0.5 \\ \vec{X}_{r_1}^t + F * (\vec{X}_{r_2}^t - \vec{X}_{r_3}^t), & rand < 0.5 \end{cases}. \quad (14)$$

In the formula, F represents the cooperation factor, its primary function is to regulate the foraging behaviour of the Arctic puffin; r_1, r_2, r_3 is a random number within the range of $[1, N - 1]$, $\vec{X}_{r_1}^t, \vec{X}_{r_2}^t, \vec{X}_{r_3}^t$ is the randomly selected candidate optimal population.

(2) Enhanced search

After searching or hunting in a specific location for a period of time, Arctic puffins will assess whether any viable resources remain there. If the remaining resources are insufficient to suit the group's demands, the Arctic puffin population will opt to hunt for another place. The position updating formula is as follows:

$$\vec{Y}_i^{t+1} = \vec{W}_i^{t+1} * (1 + f), \quad (15)$$

Table 2: The result of standard functions for different algorithms ($D = 30$).

		HHO	BOA	GTO	DBO	BWO	SABO	APO	CAAPO
F1	min	$1.00 \times 10^{+08}$	$7.88 \times 10^{+10}$	$6.20 \times 10^{+03}$	$1.05 \times 10^{+07}$	$4.13 \times 10^{+10}$	$4.92 \times 10^{+09}$	$5.71 \times 10^{+04}$	$1.56 \times 10^{+04}$
F1	Std	$2.70 \times 10^{+08}$	$9.37 \times 10^{+09}$	$3.79 \times 10^{+05}$	$2.22 \times 10^{+08}$	$4.85 \times 10^{+09}$	$4.54 \times 10^{+09}$	$2.11 \times 10^{+05}$	$1.78 \times 10^{+05}$
F1	avg	$3.81 \times 10^{+08}$	$1.07 \times 10^{+11}$	$2.34 \times 10^{+05}$	$3.02 \times 10^{+08}$	$5.33 \times 10^{+10}$	$1.38 \times 10^{+10}$	$2.22 \times 10^{+05}$	$1.58 \times 10^{+05}$
F3	min	$5.10 \times 10^{+04}$	$1.51 \times 10^{+05}$	$5.35 \times 10^{+03}$	$6.65 \times 10^{+04}$	$6.92 \times 10^{+04}$	$6.06 \times 10^{+04}$	$6.06 \times 10^{+03}$	$5.02 \times 10^{+03}$
F3	Std	$6.44 \times 10^{+03}$	$2.71 \times 10^{+04}$	$6.55 \times 10^{+03}$	$3.69 \times 10^{+04}$	$1.05 \times 10^{+04}$	$1.24 \times 10^{+04}$	$4.22 \times 10^{+03}$	$5.23 \times 10^{+03}$
F3	avg	$6.85 \times 10^{+04}$	$1.78 \times 10^{+05}$	$1.74 \times 10^{+04}$	$1.14 \times 10^{+05}$	$9.60 \times 10^{+04}$	$8.77 \times 10^{+04}$	$1.12 \times 10^{+04}$	$1.38 \times 10^{+04}$
F4	min	$5.54 \times 10^{+02}$	$2.50 \times 10^{+04}$	$4.70 \times 10^{+02}$	$4.92 \times 10^{+02}$	$7.11 \times 10^{+03}$	$1.00 \times 10^{+03}$	$4.12 \times 10^{+02}$	$4.08 \times 10^{+02}$
F4	Std	$9.36 \times 10^{+01}$	$3.93 \times 10^{+03}$	$2.96 \times 10^{+01}$	$1.33 \times 10^{+02}$	$1.67 \times 10^{+03}$	$1.06 \times 10^{+03}$	$2.26 \times 10^{+01}$	$2.73 \times 10^{+01}$
F4	avg	$6.99 \times 10^{+02}$	$3.33 \times 10^{+04}$	$5.15 \times 10^{+02}$	$7.19 \times 10^{+02}$	$1.13 \times 10^{+04}$	$2.26 \times 10^{+03}$	$4.77 \times 10^{+02}$	$4.96 \times 10^{+02}$
F5	min	$7.09 \times 10^{+02}$	$1.16 \times 10^{+03}$	$6.31 \times 10^{+02}$	$6.40 \times 10^{+02}$	$8.63 \times 10^{+02}$	$7.89 \times 10^{+02}$	$5.53 \times 10^{+02}$	$5.39 \times 10^{+02}$
F5	Std	$2.48 \times 10^{+01}$	$1.93 \times 10^{+01}$	$4.00 \times 10^{+01}$	$6.22 \times 10^{+01}$	$2.01 \times 10^{+01}$	$3.65 \times 10^{+01}$	$4.51 \times 10^{+01}$	$1.73 \times 10^{+01}$
F5	avg	$7.52 \times 10^{+02}$	$1.19 \times 10^{+03}$	$7.01 \times 10^{+02}$	$7.50 \times 10^{+02}$	$9.14 \times 10^{+02}$	$8.51 \times 10^{+02}$	$6.11 \times 10^{+02}$	$5.70 \times 10^{+02}$
F6	min	$6.50 \times 10^{+02}$	$6.93 \times 10^{+02}$	$6.24 \times 10^{+02}$	$6.23 \times 10^{+02}$	$6.74 \times 10^{+02}$	$6.35 \times 10^{+02}$	$6.01 \times 10^{+02}$	$6.00 \times 10^{+02}$
F6	Std	$5.87 \times 10^{+00}$	$5.14 \times 10^{+00}$	$7.14 \times 10^{+00}$	$1.08 \times 10^{+01}$	$4.13 \times 10^{+00}$	$1.36 \times 10^{+01}$	$1.18 \times 10^{+00}$	6.88×10^{-01}
F6	avg	$6.64 \times 10^{+02}$	$7.05 \times 10^{+02}$	$6.42 \times 10^{+02}$	$6.51 \times 10^{+02}$	$6.83 \times 10^{+02}$	$6.64 \times 10^{+02}$	$6.02 \times 10^{+02}$	$6.01 \times 10^{+02}$
F7	min	$1.28 \times 10^{+03}$	$2.16 \times 10^{+03}$	$1.01 \times 10^{+03}$	$8.81 \times 10^{+02}$	$1.51 \times 10^{+03}$	$1.06 \times 10^{+03}$	$7.98 \times 10^{+02}$	$7.79 \times 10^{+02}$
F7	std	$1.02 \times 10^{+02}$	$6.47 \times 10^{+01}$	$1.06 \times 10^{+02}$	$9.54 \times 10^{+01}$	$4.19 \times 10^{+01}$	$7.80 \times 10^{+01}$	$5.55 \times 10^{+01}$	$1.75 \times 10^{+01}$
F7	avg	$1.50 \times 10^{+03}$	$2.29 \times 10^{+03}$	$1.20 \times 10^{+03}$	$1.06 \times 10^{+03}$	$1.60 \times 10^{+03}$	$1.23 \times 10^{+03}$	$9.05 \times 10^{+02}$	$8.13 \times 10^{+02}$
F8	min	$9.73 \times 10^{+02}$	$1.45 \times 10^{+03}$	$9.09 \times 10^{+02}$	$9.20 \times 10^{+02}$	$1.16 \times 10^{+03}$	$1.06 \times 10^{+03}$	$8.41 \times 10^{+02}$	$8.31 \times 10^{+02}$
F8	std	$4.11 \times 10^{+01}$	$3.23 \times 10^{+01}$	$4.63 \times 10^{+01}$	$5.21 \times 10^{+01}$	$2.44 \times 10^{+01}$	$2.74 \times 10^{+01}$	$5.80 \times 10^{+01}$	$1.63 \times 10^{+01}$
F8	avg	$1.06 \times 10^{+03}$	$1.51 \times 10^{+03}$	$9.76 \times 10^{+02}$	$1.05 \times 10^{+03}$	$1.23 \times 10^{+03}$	$1.10 \times 10^{+03}$	$9.32 \times 10^{+02}$	$8.64 \times 10^{+02}$
F9	min	$8.65 \times 10^{+03}$	$3.22 \times 10^{+04}$	$2.90 \times 10^{+03}$	$3.28 \times 10^{+03}$	$1.14 \times 10^{+04}$	$3.65 \times 10^{+03}$	$9.28 \times 10^{+02}$	$9.25 \times 10^{+02}$
F9	std	$1.52 \times 10^{+03}$	$2.87 \times 10^{+03}$	$1.03 \times 10^{+03}$	$3.32 \times 10^{+03}$	$1.50 \times 10^{+03}$	$2.63 \times 10^{+03}$	$1.01 \times 10^{+02}$	$2.84 \times 10^{+02}$
F9	avg	$1.16 \times 10^{+04}$	$3.80 \times 10^{+04}$	$5.28 \times 10^{+03}$	$8.44 \times 10^{+03}$	$1.55 \times 10^{+04}$	$1.06 \times 10^{+04}$	$1.06 \times 10^{+03}$	$1.14 \times 10^{+03}$
F10	min	$3.81 \times 10^{+03}$	$1.50 \times 10^{+04}$	$3.85 \times 10^{+03}$	$3.46 \times 10^{+03}$	$7.86 \times 10^{+03}$	$7.85 \times 10^{+03}$	$4.27 \times 10^{+03}$	$3.47 \times 10^{+03}$
F10	std	$7.07 \times 10^{+02}$	$3.27 \times 10^{+02}$	$9.33 \times 10^{+02}$	$1.46 \times 10^{+03}$	$3.46 \times 10^{+02}$	$4.20 \times 10^{+02}$	$1.45 \times 10^{+03}$	$1.69 \times 10^{+03}$
F10	avg	$5.77 \times 10^{+03}$	$1.57 \times 10^{+04}$	$5.75 \times 10^{+03}$	$6.59 \times 10^{+03}$	$8.75 \times 10^{+03}$	$8.85 \times 10^{+03}$	$7.18 \times 10^{+03}$	$6.46 \times 10^{+03}$
F11	min	$1.31 \times 10^{+03}$	$1.93 \times 10^{+04}$	$1.18 \times 10^{+03}$	$1.34 \times 10^{+03}$	$5.33 \times 10^{+03}$	$2.53 \times 10^{+03}$	$1.14 \times 10^{+03}$	$1.13 \times 10^{+03}$
F11	std	$1.99 \times 10^{+02}$	$2.19 \times 10^{+03}$	$3.41 \times 10^{+01}$	$5.06 \times 10^{+02}$	$1.12 \times 10^{+03}$	$1.74 \times 10^{+03}$	$4.32 \times 10^{+01}$	$3.20 \times 10^{+01}$
F11	avg	$1.59 \times 10^{+03}$	$2.52 \times 10^{+04}$	$1.25 \times 10^{+03}$	$1.86 \times 10^{+03}$	$7.53 \times 10^{+03}$	$5.29 \times 10^{+03}$	$1.21 \times 10^{+03}$	$1.19 \times 10^{+03}$
F12	min	$1.03 \times 10^{+07}$	$3.43 \times 10^{+10}$	$1.22 \times 10^{+05}$	$2.25 \times 10^{+06}$	$6.29 \times 10^{+09}$	$1.26 \times 10^{+08}$	$4.35 \times 10^{+04}$	$2.73 \times 10^{+04}$
F12	std	$6.58 \times 10^{+07}$	$1.69 \times 10^{+10}$	$1.50 \times 10^{+06}$	$5.84 \times 10^{+07}$	$2.18 \times 10^{+09}$	$7.10 \times 10^{+08}$	$5.17 \times 10^{+05}$	$9.99 \times 10^{+05}$
F12	avg	$7.31 \times 10^{+07}$	$8.06 \times 10^{+10}$	$1.77 \times 10^{+06}$	$5.10 \times 10^{+07}$	$1.17 \times 10^{+10}$	$8.48 \times 10^{+08}$	$6.42 \times 10^{+05}$	$1.07 \times 10^{+06}$
F13	min	$3.85 \times 10^{+05}$	$1.35 \times 10^{+10}$	$2.97 \times 10^{+03}$	$3.26 \times 10^{+04}$	$4.34 \times 10^{+09}$	$2.03 \times 10^{+06}$	$2.30 \times 10^{+03}$	$1.46 \times 10^{+03}$
F13	std	$6.79 \times 10^{+05}$	$1.47 \times 10^{+10}$	$5.80 \times 10^{+04}$	$2.62 \times 10^{+07}$	$2.47 \times 10^{+09}$	$3.98 \times 10^{+08}$	$6.84 \times 10^{+03}$	$8.19 \times 10^{+03}$
F13	avg	$1.08 \times 10^{+06}$	$4.67 \times 10^{+10}$	$3.27 \times 10^{+04}$	$1.50 \times 10^{+07}$	$7.39 \times 10^{+09}$	$1.86 \times 10^{+08}$	$1.17 \times 10^{+04}$	$8.26 \times 10^{+03}$
F14	min	$5.80 \times 10^{+04}$	$1.12 \times 10^{+07}$	$1.65 \times 10^{+03}$	$8.98 \times 10^{+03}$	$3.38 \times 10^{+05}$	$6.10 \times 10^{+04}$	$1.47 \times 10^{+03}$	$1.46 \times 10^{+03}$
F14	std	$1.71 \times 10^{+06}$	$8.11 \times 10^{+07}$	$1.15 \times 10^{+04}$	$9.82 \times 10^{+05}$	$2.44 \times 10^{+06}$	$8.66 \times 10^{+05}$	$2.56 \times 10^{+01}$	$4.53 \times 10^{+01}$
F14	avg	$1.74 \times 10^{+06}$	$1.34 \times 10^{+08}$	$7.05 \times 10^{+03}$	$5.27 \times 10^{+05}$	$4.54 \times 10^{+06}$	$9.17 \times 10^{+05}$	$1.51 \times 10^{+03}$	$1.54 \times 10^{+03}$
F15	min	$4.88 \times 10^{+04}$	$2.74 \times 10^{+09}$	$1.71 \times 10^{+03}$	$4.42 \times 10^{+03}$	$9.08 \times 10^{+07}$	$6.53 \times 10^{+04}$	$1.82 \times 10^{+03}$	$1.57 \times 10^{+03}$
F15	std	$8.88 \times 10^{+04}$	$3.49 \times 10^{+09}$	$4.07 \times 10^{+03}$	$3.60 \times 10^{+05}$	$1.22 \times 10^{+08}$	$6.22 \times 10^{+06}$	$1.05 \times 10^{+03}$	$1.44 \times 10^{+03}$
F15	avg	$1.34 \times 10^{+05}$	$8.06 \times 10^{+09}$	$5.67 \times 10^{+03}$	$1.46 \times 10^{+05}$	$2.86 \times 10^{+08}$	$4.24 \times 10^{+06}$	$2.91 \times 10^{+03}$	$3.54 \times 10^{+03}$
F16	min	$2.56 \times 10^{+03}$	$8.19 \times 10^{+03}$	$2.32 \times 10^{+03}$	$2.74 \times 10^{+03}$	$4.69 \times 10^{+03}$	$3.76 \times 10^{+03}$	$2.10 \times 10^{+03}$	$1.74 \times 10^{+03}$
F16	std	$4.19 \times 10^{+02}$	$1.95 \times 10^{+03}$	$2.84 \times 10^{+02}$	$3.63 \times 10^{+02}$	$3.93 \times 10^{+02}$	$2.46 \times 10^{+02}$	$4.79 \times 10^{+02}$	$2.93 \times 10^{+02}$
F16	avg	$3.46 \times 10^{+03}$	$1.12 \times 10^{+04}$	$2.75 \times 10^{+03}$	$3.42 \times 10^{+03}$	$5.41 \times 10^{+03}$	$4.11 \times 10^{+03}$	$2.91 \times 10^{+03}$	$2.35 \times 10^{+03}$
F17	min	$2.15 \times 10^{+03}$	$6.25 \times 10^{+03}$	$2.01 \times 10^{+03}$	$2.07 \times 10^{+03}$	$3.01 \times 10^{+03}$	$2.37 \times 10^{+03}$	$1.77 \times 10^{+03}$	$1.74 \times 10^{+03}$
F17	std	$3.15 \times 10^{+02}$	$8.33 \times 10^{+03}$	$1.96 \times 10^{+02}$	$2.59 \times 10^{+02}$	$5.02 \times 10^{+02}$	$2.39 \times 10^{+02}$	$1.10 \times 10^{+02}$	$7.67 \times 10^{+01}$
F17	avg	$2.73 \times 10^{+03}$	$1.56 \times 10^{+04}$	$2.26 \times 10^{+03}$	$2.62 \times 10^{+03}$	$4.01 \times 10^{+03}$	$2.96 \times 10^{+03}$	$1.90 \times 10^{+03}$	$1.83 \times 10^{+03}$
F18	min	$2.77 \times 10^{+05}$	$5.88 \times 10^{+07}$	$7.53 \times 10^{+03}$	$1.57 \times 10^{+05}$	$2.03 \times 10^{+07}$	$1.16 \times 10^{+05}$	$9.42 \times 10^{+03}$	$6.51 \times 10^{+03}$
F18	std	$4.03 \times 10^{+06}$	$1.08 \times 10^{+08}$	$1.14 \times 10^{+05}$	$1.18 \times 10^{+07}$	$2.23 \times 10^{+07}$	$5.45 \times 10^{+06}$	$3.87 \times 10^{+04}$	$3.25 \times 10^{+04}$
F18	avg	$3.73 \times 10^{+06}$	$2.13 \times 10^{+08}$	$1.35 \times 10^{+05}$	$6.01 \times 10^{+06}$	$5.55 \times 10^{+07}$	$4.35 \times 10^{+06}$	$5.17 \times 10^{+04}$	$5.72 \times 10^{+04}$
F19	min	$1.28 \times 10^{+05}$	$1.01 \times 10^{+09}$	$1.98 \times 10^{+03}$	$2.51 \times 10^{+03}$	$3.47 \times 10^{+07}$	$3.14 \times 10^{+05}$	$2.07 \times 10^{+03}$	$1.95 \times 10^{+03}$
F19	std	$1.54 \times 10^{+06}$	$1.52 \times 10^{+09}$	$6.36 \times 10^{+03}$	$4.83 \times 10^{+06}$	$1.94 \times 10^{+08}$	$1.38 \times 10^{+07}$	$4.23 \times 10^{+03}$	$4.60 \times 10^{+03}$
F19	avg	$1.78 \times 10^{+06}$	$3.57 \times 10^{+09}$	$8.19 \times 10^{+03}$	$2.45 \times 10^{+06}$	$4.71 \times 10^{+08}$	$9.40 \times 10^{+06}$	$4.57 \times 10^{+03}$	$6.34 \times 10^{+03}$
F20	min	$2.51 \times 10^{+03}$	$3.99 \times 10^{+03}$	$2.26 \times 10^{+03}$	$2.30 \times 10^{+03}$	$2.86 \times 10^{+03}$	$2.78 \times 10^{+03}$	$2.04 \times 10^{+03}$	$2.04 \times 10^{+03}$
F20	std	$2.14 \times 10^{+02}$	$1.88 \times 10^{+02}$	$1.67 \times 10^{+02}$	$2.26 \times 10^{+02}$	$1.02 \times 10^{+02}$	$1.36 \times 10^{+02}$	$2.10 \times 10^{+02}$	$8.46 \times 10^{+01}$
F20	avg	$2.86 \times 10^{+03}$	$4.37 \times 10^{+03}$	$2.57 \times 10^{+03}$	$2.70 \times 10^{+03}$	$3.04 \times 10^{+03}$	$3.12 \times 10^{+03}$	$2.30 \times 10^{+03}$	$2.19 \times 10^{+03}$
F21	min	$2.48 \times 10^{+03}$	$3.10 \times 10^{+03}$	$2.39 \times 10^{+03}$	$2.48 \times 10^{+03}$	$2.66 \times 10^{+03}$	$2.55 \times 10^{+03}$	$2.34 \times 10^{+03}$	$2.33 \times 10^{+03}$
F21	std	$5.21 \times 10^{+01}$	$8.00 \times 10^{+01}$	$3.91 \times 10^{+01}$	$5.81 \times 10^{+01}$	$3.70 \times 10^{+01}$	$2.60 \times 10^{+01}$	$5.32 \times 10^{+01}$	$1.63 \times 10^{+01}$
F21	avg	$2.58 \times 10^{+03}$	$3.26 \times 10^{+03}$	$2.45 \times 10^{+03}$	$2.57 \times 10^{+03}$	$2.73 \times 10^{+03}$	$2.59 \times 10^{+03}$	$2.42 \times 10^{+03}$	$2.36 \times 10^{+03}$
F22	min	$3.23 \times 10^{+03}$	$1.46 \times 10^{+04}$	$2.30 \times 10^{+03}$	$2.37 \times 10^{+03}$	$8.29 \times 10^{+03}$	$3.08 \times 10^{+03}$	$2.30 \times 10^{+03}$	$2.30 \times 10^{+03}$
F22	std	$1.16 \times 10^{+03}$	$7.95 \times 10^{+02}$	$1.58 \times 10^{+03}$	$2.57 \times 10^{+03}$	$3.77 \times 10^{+02}$	$8.20 \times 10^{+02}$	$2.07 \times 10^{+00}$	$2.01 \times 10^{+00}$
F22	avg	$7.51 \times 10^{+03}$	$1.71 \times 10^{+04}$	$2.83 \times 10^{+03}$	$4.74 \times 10^{+03}$	$8.91 \times 10^{+03}$	$4.38 \times 10^{+03}$	$2.31 \times 10^{+03}$	$2.30 \times 10^{+03}$

Table 2: Continued

		HHO	BOA	GTO	DBO	BWO	SABO	APO	CAAPO
F23	min	$3.00 \times 10^{+03}$	$4.19 \times 10^{+03}$	$2.77 \times 10^{+03}$	$2.85 \times 10^{+03}$	$3.19 \times 10^{+03}$	$2.97 \times 10^{+03}$	$2.70 \times 10^{+03}$	$2.67 \times 10^{+03}$
F23	std	$1.40 \times 10^{+02}$	$1.96 \times 10^{+02}$	$6.75 \times 10^{+01}$	$9.78 \times 10^{+01}$	$5.78 \times 10^{+01}$	$1.00 \times 10^{+02}$	$4.99 \times 10^{+01}$	$2.11 \times 10^{+01}$
F23	avg	$3.27 \times 10^{+03}$	$4.69 \times 10^{+03}$	$2.86 \times 10^{+03}$	$3.01 \times 10^{+03}$	$3.33 \times 10^{+03}$	$3.15 \times 10^{+03}$	$2.78 \times 10^{+03}$	$2.71 \times 10^{+03}$
F24	min	$3.22 \times 10^{+03}$	$5.04 \times 10^{+03}$	$2.92 \times 10^{+03}$	$2.99 \times 10^{+03}$	$3.45 \times 10^{+03}$	$3.05 \times 10^{+03}$	$2.86 \times 10^{+03}$	$2.84 \times 10^{+03}$
F24	std	$1.50 \times 10^{+02}$	$3.35 \times 10^{+02}$	$7.87 \times 10^{+01}$	$9.14 \times 10^{+01}$	$8.52 \times 10^{+01}$	$1.16 \times 10^{+02}$	$4.39 \times 10^{+01}$	$2.05 \times 10^{+01}$
F24	avg	$3.55 \times 10^{+03}$	$5.58 \times 10^{+03}$	$3.04 \times 10^{+03}$	$3.18 \times 10^{+03}$	$3.58 \times 10^{+03}$	$3.25 \times 10^{+03}$	$2.92 \times 10^{+03}$	$2.88 \times 10^{+03}$
F25	min	$2.95 \times 10^{+03}$	$1.30 \times 10^{+04}$	$2.89 \times 10^{+03}$	$2.90 \times 10^{+03}$	$4.15 \times 10^{+03}$	$3.23 \times 10^{+03}$	$2.89 \times 10^{+03}$	$2.88 \times 10^{+03}$
F25	std	$3.90 \times 10^{+01}$	$1.21 \times 10^{+03}$	$2.09 \times 10^{+01}$	$1.90 \times 10^{+02}$	$1.46 \times 10^{+02}$	$1.44 \times 10^{+02}$	$1.07 \times 10^{+01}$	$1.92 \times 10^{+01}$
F25	avg	$3.00 \times 10^{+03}$	$1.59 \times 10^{+04}$	$2.92 \times 10^{+03}$	$3.02 \times 10^{+03}$	$4.49 \times 10^{+03}$	$3.42 \times 10^{+03}$	$2.90 \times 10^{+03}$	$2.91 \times 10^{+03}$
F26	min	$6.88 \times 10^{+03}$	$1.58 \times 10^{+04}$	$2.81 \times 10^{+03}$	$5.56 \times 10^{+03}$	$1.02 \times 10^{+04}$	$6.01 \times 10^{+03}$	$2.88 \times 10^{+03}$	$2.81 \times 10^{+03}$
F26	std	$9.53 \times 10^{+02}$	$6.39 \times 10^{+02}$	$1.38 \times 10^{+03}$	$8.53 \times 10^{+02}$	$4.15 \times 10^{+02}$	$8.22 \times 10^{+02}$	$8.47 \times 10^{+02}$	$7.67 \times 10^{+02}$
F26	avg	$8.38 \times 10^{+03}$	$1.75 \times 10^{+04}$	$5.69 \times 10^{+03}$	$7.25 \times 10^{+03}$	$1.09 \times 10^{+04}$	$8.26 \times 10^{+03}$	$4.43 \times 10^{+03}$	$3.56 \times 10^{+03}$
F27	min	$3.32 \times 10^{+03}$	$5.33 \times 10^{+03}$	$3.22 \times 10^{+03}$	$3.25 \times 10^{+03}$	$3.73 \times 10^{+03}$	$3.28 \times 10^{+03}$	$3.22 \times 10^{+03}$	$3.21 \times 10^{+03}$
F27	std	$1.81 \times 10^{+02}$	$8.81 \times 10^{+02}$	$6.54 \times 10^{+01}$	$6.02 \times 10^{+01}$	$1.68 \times 10^{+02}$	$1.34 \times 10^{+02}$	$1.45 \times 10^{+01}$	$1.36 \times 10^{+01}$
F27	avg	$3.60 \times 10^{+03}$	$6.89 \times 10^{+03}$	$3.31 \times 10^{+03}$	$3.34 \times 10^{+03}$	$4.07 \times 10^{+03}$	$3.48 \times 10^{+03}$	$3.26 \times 10^{+03}$	$3.24 \times 10^{+03}$
F28	min	$3.34 \times 10^{+03}$	$1.18 \times 10^{+04}$	$3.20 \times 10^{+03}$	$3.30 \times 10^{+03}$	$5.61 \times 10^{+03}$	$3.72 \times 10^{+03}$	$3.21 \times 10^{+03}$	$3.21 \times 10^{+03}$
F28	std	$9.19 \times 10^{+01}$	$1.40 \times 10^{+03}$	$4.16 \times 10^{+01}$	$5.73 \times 10^{+02}$	$3.24 \times 10^{+02}$	$4.29 \times 10^{+02}$	$1.75 \times 10^{+01}$	$2.68 \times 10^{+01}$
F28	avg	$3.47 \times 10^{+03}$	$1.48 \times 10^{+04}$	$3.29 \times 10^{+03}$	$3.58 \times 10^{+03}$	$6.36 \times 10^{+03}$	$4.37 \times 10^{+03}$	$3.23 \times 10^{+03}$	$3.25 \times 10^{+03}$
F29	min	$4.35 \times 10^{+03}$	$3.32 \times 10^{+04}$	$3.67 \times 10^{+03}$	$3.87 \times 10^{+03}$	$5.28 \times 10^{+03}$	$4.77 \times 10^{+03}$	$3.42 \times 10^{+03}$	$3.36 \times 10^{+03}$
F29	std	$6.11 \times 10^{+02}$	$1.82 \times 10^{+05}$	$4.08 \times 10^{+02}$	$3.35 \times 10^{+02}$	$9.01 \times 10^{+02}$	$7.19 \times 10^{+02}$	$1.38 \times 10^{+02}$	$9.38 \times 10^{+01}$
F29	avg	$5.11 \times 10^{+03}$	$2.24 \times 10^{+05}$	$4.35 \times 10^{+03}$	$4.50 \times 10^{+03}$	$7.08 \times 10^{+03}$	$5.99 \times 10^{+03}$	$3.65 \times 10^{+03}$	$3.52 \times 10^{+03}$
F30	min	$1.27 \times 10^{+06}$	$1.63 \times 10^{+09}$	$9.15 \times 10^{+03}$	$3.97 \times 10^{+04}$	$2.33 \times 10^{+08}$	$4.81 \times 10^{+06}$	$8.47 \times 10^{+03}$	$6.63 \times 10^{+03}$
F30	std	$8.40 \times 10^{+06}$	$3.22 \times 10^{+09}$	$1.41 \times 10^{+04}$	$7.61 \times 10^{+06}$	$4.78 \times 10^{+08}$	$4.63 \times 10^{+07}$	$1.06 \times 10^{+04}$	$7.51 \times 10^{+03}$
F30	avg	$1.02 \times 10^{+07}$	$7.94 \times 10^{+09}$	$2.30 \times 10^{+04}$	$4.53 \times 10^{+06}$	$1.06 \times 10^{+09}$	$4.09 \times 10^{+07}$	$2.35 \times 10^{+04}$	$1.38 \times 10^{+04}$

$$f = 0.1 * (rand - 1) * \frac{T - t}{T}. \quad (16)$$

In the formula: T represents the total number of iterations; t represents the current iteration number; $rand$ represents the random number.

(3) Avoiding predators

When Arctic puffins are hunting or foraging, if they detect the presence of enemies or predators nearby, they will emit a call to alert other individuals in the group and change their position to fly to a safe area for refuge. The position update formula is as follows:

$$\vec{Z}_i^{t+1} = \begin{cases} \vec{X}_i^t + F * L(D) + (\vec{X}_{r_1}^t - \vec{X}_{r_2}^t), & rand \geq 0.5 \\ \vec{X}_i^t + \beta * (\vec{X}_{r_2}^t - \vec{X}_{r_3}^t), & rand < 0.5 \end{cases}. \quad (17)$$

In the formula, β is a random number between $[0,1]$.

During the underwater predation phase, the Arctic puffin algorithm updates locations in various stages. The following formula is applied for merging to prevent conflicts or inaccuracies between the optimal populations and optimal solutions of the three sub-phases:

$$\vec{P}_i^{t+1} = \vec{W}_i^{t+1} \cup \vec{Y}_i^{t+1} \cup \vec{Z}_i^{t+1}, \quad (18)$$

$$new = \text{sort}(\vec{P}_i^{t+1}), \quad (19)$$

$$\vec{X}_i^{t+1} = new(1 : N), \quad (20)$$

In APO, in order to better achieve the transition from the aerial search phase to the foraging phase, the transition coefficient B and specific parameter C are introduced here:

$$B = 2 * \log(1/rand) * (1 - t/T). \quad (21)$$

4. Improve the Arctic Puffin Algorithm

4.1. Tent chaotic mapping based on random variables

Uneven population distribution is an inevitable consequence of the APO algorithm's random population initialization. According to the APO algorithm's search efficacy, the likelihood of discovering the best answer increases with the population's level of individual distribution (Akraam et al., 2023; Al-Daraiseh et al., 2023). Consequently, the population's initial distribution is quite important. The field of population initialization extensively employs chaotic mapping (Gezici & Livatyali, 2022). The Tent chaotic map features strong ergodic properties and a straightforward mathematical formulation (Kanwal et al., 2022). This is expressed as

$$Y_{i+1} = \begin{cases} \frac{Y_i}{a}, & 0 \leq Y_i < a \\ \frac{1-Y_i}{1-a}, & a \leq Y_i \leq 1 \end{cases}. \quad (22)$$

In the Equation 22, Y_i and Y_{i+1} represent the chaotic values of the i th and $(i + 1)$ th iterations, respectively. When $a \in [0,1]$ and $Y_i \in [0,1]$, the system is in a chaotic state. Set a to 0.3, the initial value Y to 0.32, and the number of iterations to 500. It can be seen that the Tent map is relatively uniformly distributed within the interval $[0,1]$. The Tent chaotic map sequence within the interval $[0,1]$ is shown in Figure 2.

The chaotic sequences generated by the Tent map have good distribution and randomness. By selecting multiple different initial values to generate chaotic sequences within $[0,1]$, these sequences can replace the random values used for population initialization in the APO algorithm, as shown in Equation 23, thereby completing the population initialization process:

$$X_i^{new} = (ub_i - lb_i)Y_{i+1} + lb_i. \quad (23)$$

In the Equation: ub_i and lb_i are the upper and lower bounds of the objective function variables; X_i^{new} are the positions of the Arctic puffin population updated using the Tent chaotic map.

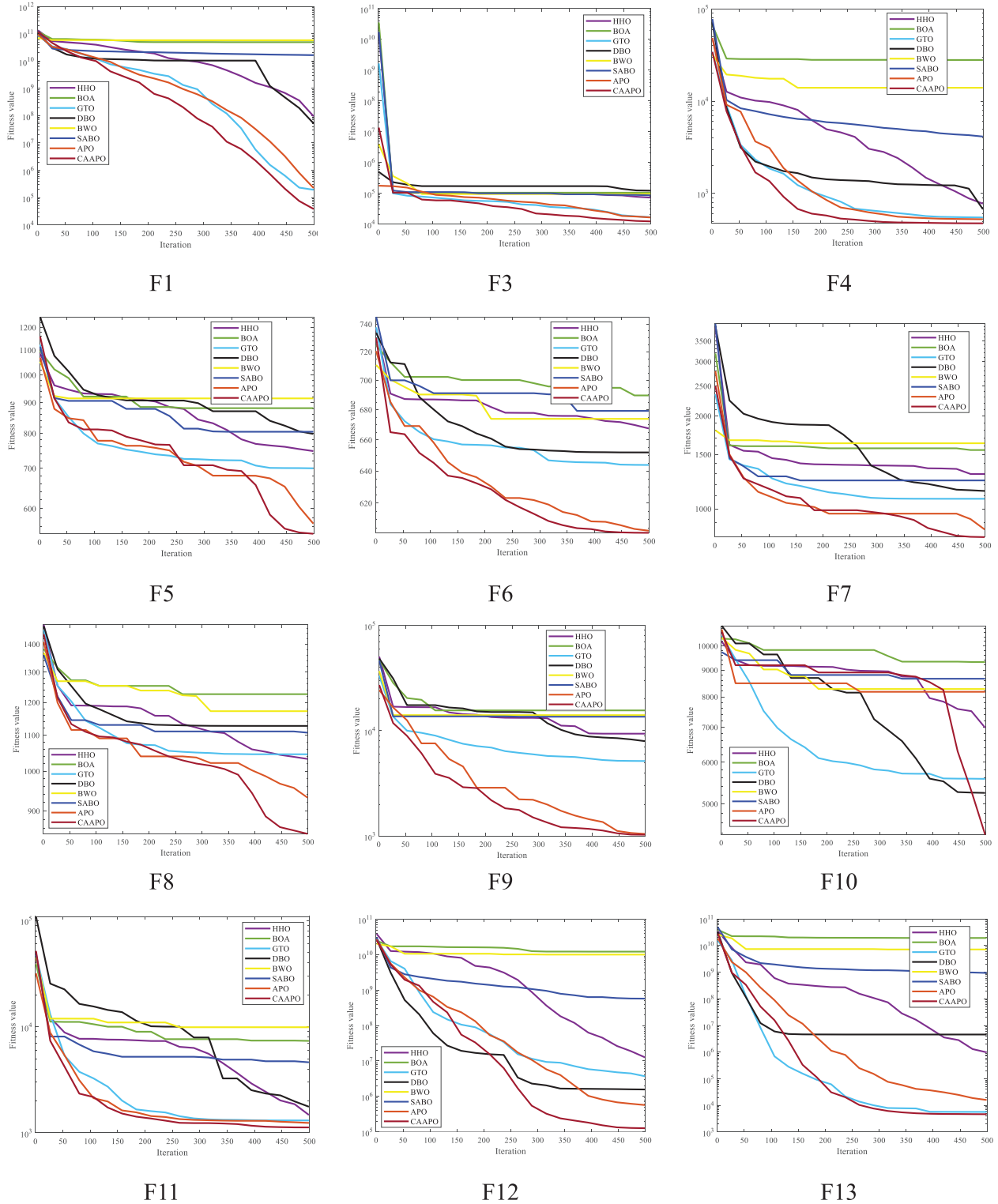


Figure 4: Iteration profile of different improved algorithms($D = 30$).

4.2. Lens imaging inverse learning strategy

This study employs the Arctic puffin's flight behaviour combined with the optical lens imaging principle (Ouyang et al., 2021; Zhang & Chen, 2024). Assume that P represents a person in the $[l_b, u_b]$ range, and that h is its height. The global best solution is repre-

sented by X , which is its projection on the X -axis. Place a convex lens with a focal length equal to F at point O , which is the midpoint of $[l_b, u_b]$. A new inverted picture P_* with a size of h_* is created when an item P is processed through the convex lens. The following conclusions may be made using the conventional convex lens

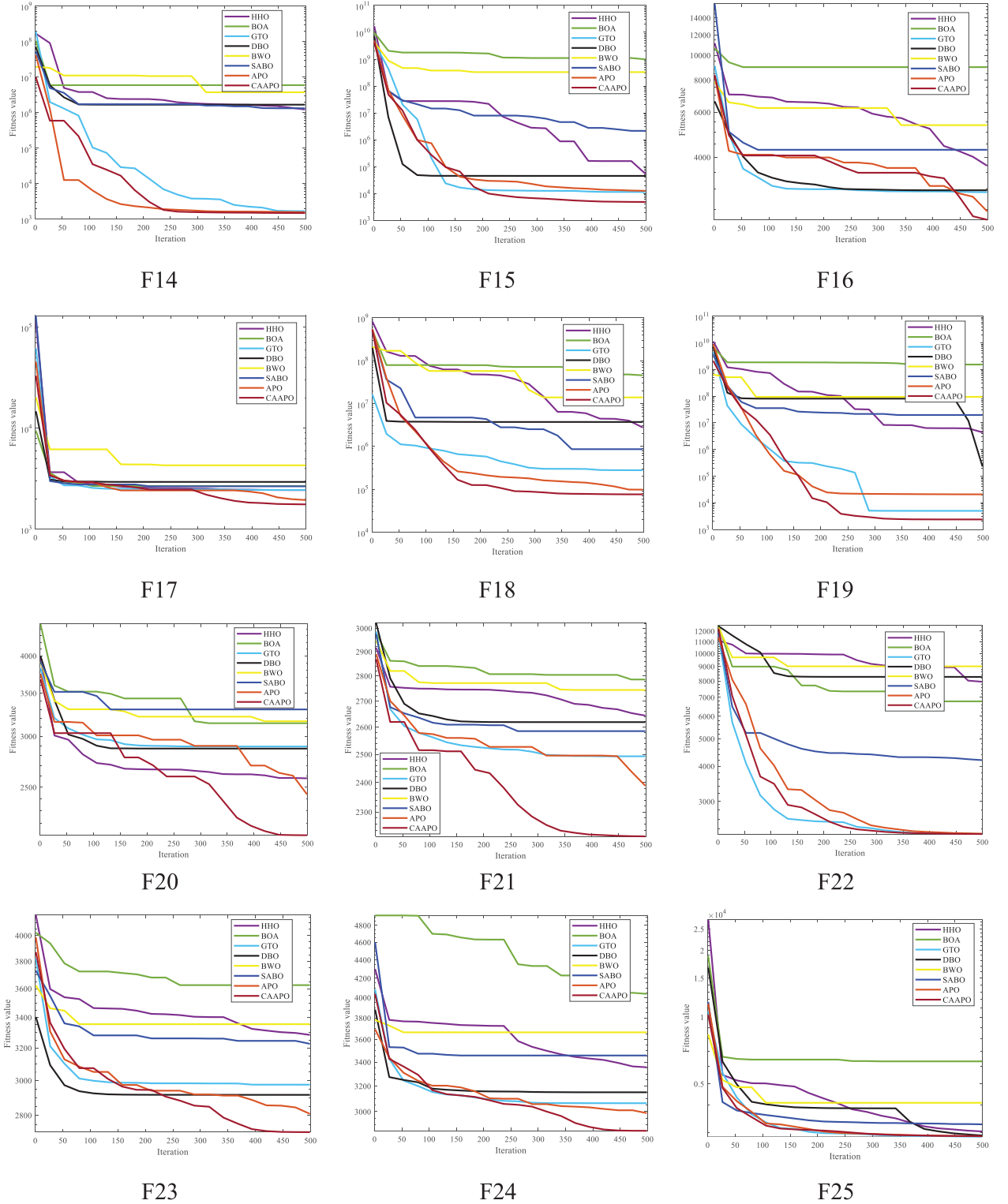


Figure 4: Continued.

imaging principles:

$$\frac{(ub + lb)/2 - X}{X^* - (ub + lb)/2} = \eta. \quad (24)$$

In Equation 24, $\eta = \frac{h}{h^*}$, is called the scaling factor. The inverse point can be obtained from Equation 24:

$$X^* = \frac{ub + lb}{2} + \frac{ub + lb}{2\eta} - \frac{X}{\eta}. \quad (25)$$

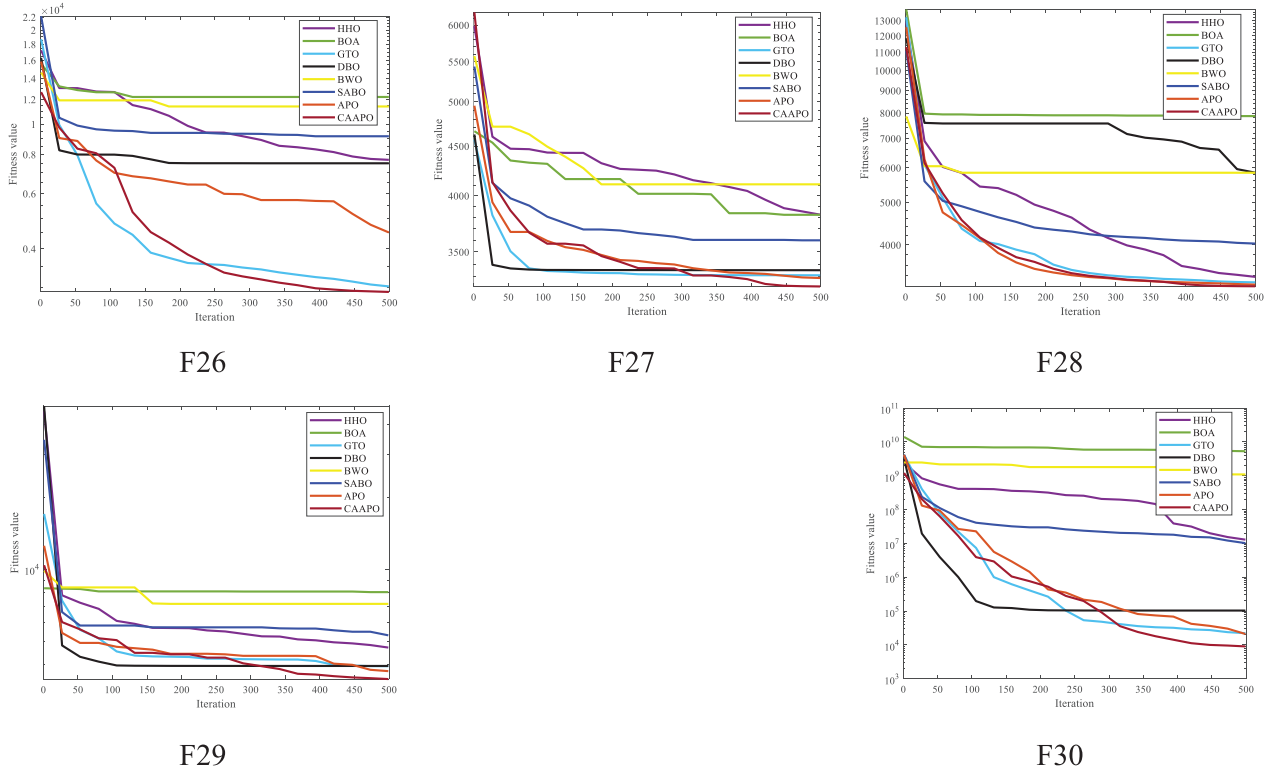


Figure 4: Continued.

After extending Equation 25 to the D -dimensional search space, the corresponding results are obtained:

$$X_j^* = ub_j + lb_j - X_j. \quad (26)$$

In Equation 26: X_j and X_j^* are the j th dimensional vectors of X and X^* , respectively, and ub_j and lb_j are the j th dimensional vectors of the decision variables, respectively. At the same time, this paper introduces a greedy mechanism to select individuals after reverse learning, thereby obtaining the optimal individual. The mathematical model of the greedy mechanism is shown in Equation 27:

$$X_{new}(t) = \begin{cases} X^*, & f(X) \geq f(X^*) \\ X, & f(X) < f(X^*) \end{cases}. \quad (27)$$

4.3. Adaptive inertia weighting strategy

The Arctic puffin's capacity to inherit its predecessor's position is indicated by the inertia weight ω . Exploration capabilities are improved when the inertia weight is high (Cao et al., 2023). The ability to exploit is enhanced when the inertia weight is low. The i th Arctic puffin's location will be changed depending on the positions of two to three randomly chosen Arctic puffins, as can be observed from the underwater foraging update formulae (9–16). It is prone to becoming trapped in local optima due to its significant reliance on other individuals (Jena & Satapathy, 2024). The extent to which past individuals impact the current individual is determined by a linearly decreasing inertia weight, which is incorporated to balance better the Arctic puffin algorithm's exploration and exploitation capabilities. The Arctic puffin's new formula for underwater foraging positions is

Formula for updating the foraging position:

$$\vec{W}_i^{t+1} = \begin{cases} \vec{X}_{r_1}^t + \omega * F * L(D) + (\vec{X}_{r_2}^t - \vec{X}_{r_3}^t), & rand \geq 0.5 \\ \vec{X}_{r_1}^t + \omega * F * (\vec{X}_{r_2}^t - \vec{X}_{r_3}^t), & rand < 0.5 \end{cases}. \quad (28)$$

Formula for updating the position to avoid predators:

$$\vec{Z}_i^{t+1} = \begin{cases} \vec{X}_i^t + \omega * F * L(D) + (\vec{X}_{r_1}^t - \vec{X}_{r_2}^t), & rand \geq 0.5 \\ \vec{X}_i^t + \beta * (\vec{X}_{r_2}^t - \vec{X}_{r_3}^t), & rand < 0.5 \end{cases}. \quad (29)$$

$$\omega = (\omega_s - \omega_e)(T - t)/T + \omega_e, \quad (30)$$

where ω_s is the initial inertia weight, ω_e is the inertia weight when the maximum number of iterations is reached, t is the current iteration number, and T is the maximum number of iterations. Inertia weight $\omega_s = 1$, $\omega_e = 0.4$, the algorithm has optimal performance. As the iterations progress, the inertia weight decreases linearly from 1 to 0.4. A larger inertia weight during initial iterations enhances the algorithm's exploration capability, while a smaller inertia weight in later iterations improves its exploitation capability.

4.4. Algorithm process

The pseudocode for CAAPO is as follows, and the detailed flowchart is shown in Figure 3: initialize parameters T , N , D , etc., initialize population X , and calculate fitness values.

4.5. Algorithm complexity analysis

The time complexity of the APO algorithm is primarily determined by three computational components: population initialization ($O(N \cdot D)$), fitness evaluation ($O(N \cdot D \cdot T)$), and iterative population updates ($O(N \cdot D \cdot T)$). The aggregate complexity therefore scales as $O(N \cdot D \cdot T)$, where N denotes population size, D problem dimensionality, and T maximum iterations. Algorithmic enhancements

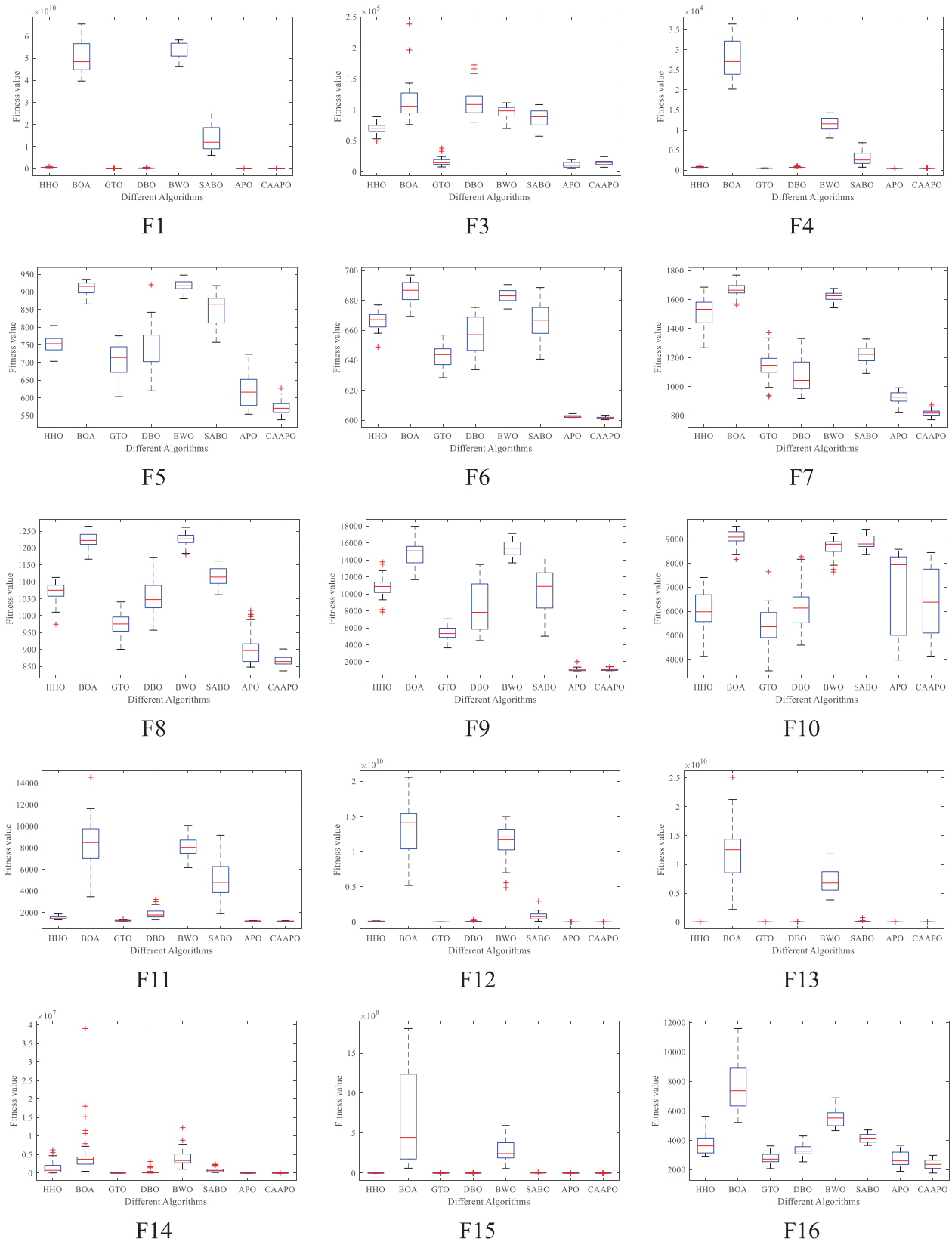


Figure 5: Box plots of iterative data distributions($D = 30$).

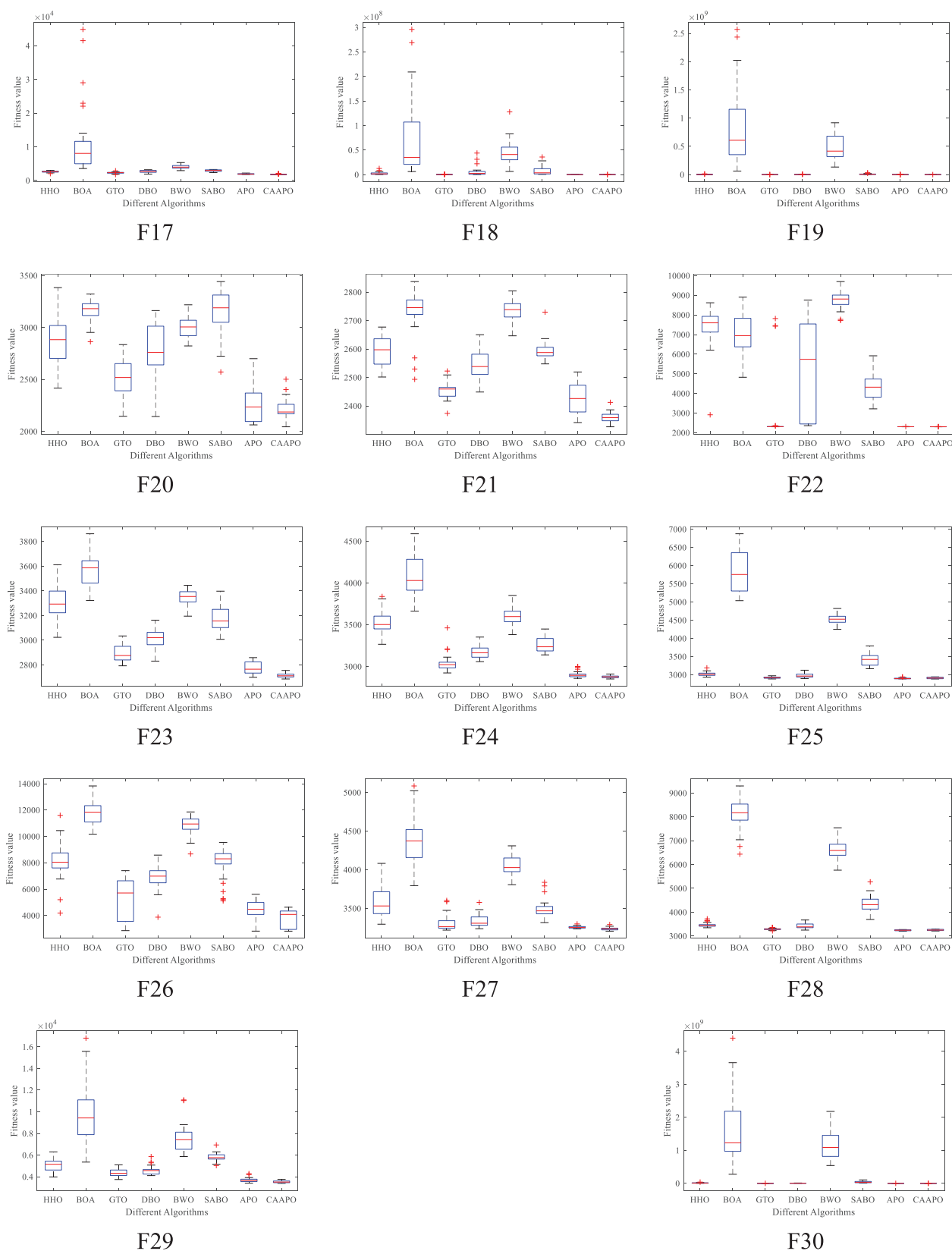


Figure 5: Continued.

Table 3: The result of standard functions for different algorithms ($D = 50$).

		HHO	BOA	GTO	DBO	BWO	SABO	APO	CAAPO
F1	min	$2.76 \times 10^{+09}$	$8.71 \times 10^{+10}$	$2.41 \times 10^{+07}$	$9.14 \times 10^{+08}$	$9.82 \times 10^{+10}$	$2.78 \times 10^{+10}$	$5.10 \times 10^{+07}$	$1.79 \times 10^{+07}$
F1	std	$1.43 \times 10^{+09}$	$8.68 \times 10^{+09}$	$2.19 \times 10^{+08}$	$7.46 \times 10^{+09}$	$4.08 \times 10^{+09}$	$1.00 \times 10^{+10}$	$5.55 \times 10^{+07}$	$5.11 \times 10^{+07}$
F1	avg	$4.91 \times 10^{+09}$	$1.04 \times 10^{+11}$	$1.81 \times 10^{+08}$	$6.58 \times 10^{+09}$	$1.06 \times 10^{+11}$	$4.65 \times 10^{+10}$	$1.22 \times 10^{+08}$	$8.39 \times 10^{+07}$
F3	min	$1.08 \times 10^{+05}$	$1.47 \times 10^{+05}$	$4.40 \times 10^{+04}$	$1.40 \times 10^{+05}$	$1.30 \times 10^{+05}$	$1.22 \times 10^{+05}$	$3.94 \times 10^{+04}$	$4.24 \times 10^{+04}$
F3	std	$1.02 \times 10^{+04}$	$4.17 \times 10^{+04}$	$9.70 \times 10^{+03}$	$7.33 \times 10^{+04}$	$1.03 \times 10^{+04}$	$1.10 \times 10^{+04}$	$1.05 \times 10^{+04}$	$1.01 \times 10^{+04}$
F3	avg	$1.25 \times 10^{+05}$	$1.82 \times 10^{+05}$	$6.45 \times 10^{+04}$	$2.13 \times 10^{+05}$	$1.48 \times 10^{+05}$	$1.45 \times 10^{+05}$	$6.16 \times 10^{+04}$	$6.65 \times 10^{+04}$
F4	min	$9.56 \times 10^{+02}$	$2.44 \times 10^{+04}$	$6.02 \times 10^{+02}$	$8.12 \times 10^{+02}$	$1.69 \times 10^{+04}$	$2.61 \times 10^{+03}$	$6.17 \times 10^{+02}$	$5.83 \times 10^{+02}$
F4	std	$2.51 \times 10^{+02}$	$4.98 \times 10^{+03}$	$8.52 \times 10^{+01}$	$3.36 \times 10^{+02}$	$2.29 \times 10^{+03}$	$2.12 \times 10^{+03}$	$3.73 \times 10^{+01}$	$4.28 \times 10^{+01}$
F4	avg	$1.44 \times 10^{+03}$	$3.41 \times 10^{+04}$	$7.17 \times 10^{+02}$	$1.15 \times 10^{+03}$	$2.19 \times 10^{+04}$	$6.30 \times 10^{+03}$	$6.72 \times 10^{+02}$	$6.66 \times 10^{+02}$
F5	min	$8.79 \times 10^{+02}$	$1.13 \times 10^{+03}$	$7.75 \times 10^{+02}$	$7.88 \times 10^{+02}$	$1.14 \times 10^{+03}$	$1.03 \times 10^{+03}$	$6.54 \times 10^{+02}$	$6.37 \times 10^{+02}$
F5	std	$3.65 \times 10^{+01}$	$2.39 \times 10^{+01}$	$3.35 \times 10^{+01}$	$9.59 \times 10^{+01}$	$2.32 \times 10^{+01}$	$4.41 \times 10^{+01}$	$8.57 \times 10^{+01}$	$3.03 \times 10^{+01}$
F5	avg	$9.34 \times 10^{+02}$	$1.20 \times 10^{+03}$	$8.51 \times 10^{+02}$	$9.79 \times 10^{+02}$	$1.20 \times 10^{+03}$	$1.13 \times 10^{+03}$	$8.26 \times 10^{+02}$	$7.02 \times 10^{+02}$
F6	min	$6.70 \times 10^{+02}$	$6.95 \times 10^{+02}$	$6.46 \times 10^{+02}$	$6.38 \times 10^{+02}$	$6.88 \times 10^{+02}$	$6.62 \times 10^{+02}$	$6.09 \times 10^{+02}$	$6.05 \times 10^{+02}$
F6	std	$3.54 \times 10^{+00}$	$3.97 \times 10^{+00}$	$6.36 \times 10^{+00}$	$1.53 \times 10^{+01}$	$5.25 \times 10^{+00}$	$1.15 \times 10^{+01}$	$3.09 \times 10^{+00}$	$2.94 \times 10^{+00}$
F6	avg	$6.78 \times 10^{+02}$	$7.03 \times 10^{+02}$	$6.56 \times 10^{+02}$	$6.65 \times 10^{+02}$	$7.00 \times 10^{+02}$	$6.87 \times 10^{+02}$	$6.14 \times 10^{+02}$	$6.09 \times 10^{+02}$
F7	min	$1.93 \times 10^{+03}$	$2.11 \times 10^{+03}$	$1.47 \times 10^{+03}$	$1.27 \times 10^{+03}$	$2.19 \times 10^{+03}$	$1.64 \times 10^{+03}$	$1.01 \times 10^{+03}$	$9.42 \times 10^{+02}$
F7	std	$1.09 \times 10^{+02}$	$7.14 \times 10^{+01}$	$1.48 \times 10^{+02}$	$1.52 \times 10^{+02}$	$4.43 \times 10^{+01}$	$8.61 \times 10^{+01}$	$7.53 \times 10^{+01}$	$4.17 \times 10^{+01}$
F7	avg	$2.15 \times 10^{+03}$	$2.31 \times 10^{+03}$	$1.72 \times 10^{+03}$	$1.50 \times 10^{+03}$	$2.29 \times 10^{+03}$	$1.79 \times 10^{+03}$	$1.21 \times 10^{+03}$	$1.02 \times 10^{+03}$
F8	min	$1.21 \times 10^{+03}$	$1.46 \times 10^{+03}$	$1.07 \times 10^{+03}$	$1.10 \times 10^{+03}$	$1.46 \times 10^{+03}$	$1.34 \times 10^{+03}$	$9.76 \times 10^{+02}$	$9.47 \times 10^{+02}$
F8	std	$3.15 \times 10^{+01}$	$2.15 \times 10^{+01}$	$3.93 \times 10^{+01}$	$8.68 \times 10^{+01}$	$2.09 \times 10^{+01}$	$5.63 \times 10^{+01}$	$7.01 \times 10^{+01}$	$4.15 \times 10^{+01}$
F8	avg	$1.27 \times 10^{+03}$	$1.52 \times 10^{+03}$	$1.17 \times 10^{+03}$	$1.30 \times 10^{+03}$	$1.52 \times 10^{+03}$	$1.45 \times 10^{+03}$	$1.09 \times 10^{+03}$	$1.02 \times 10^{+03}$
F9	min	$2.36 \times 10^{+04}$	$3.00 \times 10^{+04}$	$9.73 \times 10^{+03}$	$8.77 \times 10^{+03}$	$3.07 \times 10^{+04}$	$1.50 \times 10^{+04}$	$2.73 \times 10^{+03}$	$1.93 \times 10^{+03}$
F9	std	$3.36 \times 10^{+03}$	$3.47 \times 10^{+03}$	$1.07 \times 10^{+03}$	$1.43 \times 10^{+03}$	$2.88 \times 10^{+03}$	$5.99 \times 10^{+03}$	$1.42 \times 10^{+03}$	$2.18 \times 10^{+03}$
F9	avg	$3.04 \times 10^{+04}$	$3.90 \times 10^{+04}$	$1.24 \times 10^{+04}$	$1.25 \times 10^{+04}$	$3.75 \times 10^{+04}$	$3.16 \times 10^{+04}$	$5.01 \times 10^{+03}$	$4.74 \times 10^{+03}$
F10	min	$8.35 \times 10^{+03}$	$1.46 \times 10^{+04}$	$7.34 \times 10^{+03}$	$7.51 \times 10^{+03}$	$1.45 \times 10^{+04}$	$1.39 \times 10^{+04}$	$7.90 \times 10^{+03}$	$6.50 \times 10^{+03}$
F10	std	$1.08 \times 10^{+03}$	$4.64 \times 10^{+02}$	$2.07 \times 10^{+03}$	$2.46 \times 10^{+03}$	$3.87 \times 10^{+02}$	$4.46 \times 10^{+02}$	$2.27 \times 10^{+03}$	$2.25 \times 10^{+03}$
F10	avg	$1.05 \times 10^{+04}$	$1.57 \times 10^{+04}$	$9.70 \times 10^{+03}$	$1.14 \times 10^{+04}$	$1.51 \times 10^{+04}$	$1.49 \times 10^{+04}$	$1.32 \times 10^{+04}$	$9.34 \times 10^{+03}$
F11	min	$1.93 \times 10^{+03}$	$2.06 \times 10^{+04}$	$1.34 \times 10^{+03}$	$2.18 \times 10^{+03}$	$1.72 \times 10^{+04}$	$6.63 \times 10^{+03}$	$1.36 \times 10^{+03}$	$1.28 \times 10^{+03}$
F11	std	$8.28 \times 10^{+02}$	$2.29 \times 10^{+03}$	$1.11 \times 10^{+02}$	$2.20 \times 10^{+03}$	$1.77 \times 10^{+03}$	$2.54 \times 10^{+03}$	$2.11 \times 10^{+02}$	$2.19 \times 10^{+02}$
F11	avg	$3.39 \times 10^{+03}$	$2.52 \times 10^{+04}$	$1.50 \times 10^{+03}$	$4.40 \times 10^{+03}$	$2.26 \times 10^{+04}$	$1.07 \times 10^{+04}$	$1.64 \times 10^{+03}$	$1.50 \times 10^{+03}$
F12	min	$1.09 \times 10^{+08}$	$4.81 \times 10^{+10}$	$3.22 \times 10^{+06}$	$1.51 \times 10^{+08}$	$3.54 \times 10^{+10}$	$2.97 \times 10^{+09}$	$3.68 \times 10^{+06}$	$3.15 \times 10^{+06}$
F12	std	$6.66 \times 10^{+08}$	$1.48 \times 10^{+10}$	$2.44 \times 10^{+07}$	$8.64 \times 10^{+08}$	$9.64 \times 10^{+09}$	$6.02 \times 10^{+09}$	$5.32 \times 10^{+06}$	$5.09 \times 10^{+06}$
F12	avg	$8.47 \times 10^{+08}$	$7.38 \times 10^{+10}$	$2.65 \times 10^{+07}$	$1.02 \times 10^{+09}$	$6.73 \times 10^{+10}$	$1.18 \times 10^{+10}$	$9.74 \times 10^{+06}$	$9.25 \times 10^{+06}$
F13	min	$6.90 \times 10^{+06}$	$1.89 \times 10^{+10}$	$1.02 \times 10^{+04}$	$5.42 \times 10^{+04}$	$1.20 \times 10^{+10}$	$2.02 \times 10^{+08}$	$3.25 \times 10^{+03}$	$2.12 \times 10^{+03}$
F13	std	$2.10 \times 10^{+07}$	$1.32 \times 10^{+10}$	$3.53 \times 10^{+04}$	$1.21 \times 10^{+08}$	$9.34 \times 10^{+09}$	$2.67 \times 10^{+09}$	$3.63 \times 10^{+03}$	$2.25 \times 10^{+03}$
F13	avg	$2.85 \times 10^{+07}$	$4.28 \times 10^{+10}$	$3.52 \times 10^{+04}$	$7.35 \times 10^{+07}$	$3.65 \times 10^{+10}$	$2.30 \times 10^{+09}$	$6.82 \times 10^{+03}$	$4.97 \times 10^{+03}$
F14	min	$5.78 \times 10^{+05}$	$5.95 \times 10^{+06}$	$1.61 \times 10^{+04}$	$3.80 \times 10^{+05}$	$2.58 \times 10^{+07}$	$6.02 \times 10^{+05}$	$3.81 \times 10^{+03}$	$2.46 \times 10^{+03}$
F14	std	$3.56 \times 10^{+06}$	$1.02 \times 10^{+08}$	$9.81 \times 10^{+04}$	$3.94 \times 10^{+06}$	$2.05 \times 10^{+07}$	$5.36 \times 10^{+06}$	$3.80 \times 10^{+04}$	$5.30 \times 10^{+04}$
F14	avg	$4.98 \times 10^{+06}$	$1.47 \times 10^{+08}$	$1.07 \times 10^{+05}$	$3.90 \times 10^{+06}$	$6.67 \times 10^{+07}$	$7.40 \times 10^{+06}$	$3.45 \times 10^{+04}$	$6.30 \times 10^{+04}$
F15	min	$3.89 \times 10^{+05}$	$2.20 \times 10^{+09}$	$3.44 \times 10^{+03}$	$3.22 \times 10^{+04}$	$3.66 \times 10^{+09}$	$1.49 \times 10^{+07}$	$2.06 \times 10^{+03}$	$1.95 \times 10^{+03}$
F15	std	$6.63 \times 10^{+05}$	$3.11 \times 10^{+09}$	$1.05 \times 10^{+04}$	$1.80 \times 10^{+07}$	$1.46 \times 10^{+09}$	$4.17 \times 10^{+08}$	$3.19 \times 10^{+03}$	$2.04 \times 10^{+03}$
F15	avg	$1.32 \times 10^{+06}$	$9.26 \times 10^{+09}$	$2.05 \times 10^{+04}$	$7.44 \times 10^{+06}$	$5.96 \times 10^{+09}$	$3.60 \times 10^{+08}$	$6.09 \times 10^{+03}$	$3.87 \times 10^{+03}$
F16	min	$3.36 \times 10^{+03}$	$8.44 \times 10^{+03}$	$2.66 \times 10^{+03}$	$3.46 \times 10^{+03}$	$6.54 \times 10^{+03}$	$5.32 \times 10^{+03}$	$2.54 \times 10^{+03}$	$2.30 \times 10^{+03}$
F16	std	$7.97 \times 10^{+02}$	$1.87 \times 10^{+03}$	$5.84 \times 10^{+02}$	$6.91 \times 10^{+02}$	$9.77 \times 10^{+02}$	$4.66 \times 10^{+02}$	$7.60 \times 10^{+02}$	$3.98 \times 10^{+02}$
F16	avg	$5.22 \times 10^{+03}$	$1.11 \times 10^{+04}$	$3.76 \times 10^{+03}$	$4.83 \times 10^{+03}$	$8.96 \times 10^{+03}$	$6.09 \times 10^{+03}$	$3.76 \times 10^{+03}$	$2.98 \times 10^{+03}$
F17	min	$3.01 \times 10^{+03}$	$6.03 \times 10^{+03}$	$2.70 \times 10^{+03}$	$3.28 \times 10^{+03}$	$4.82 \times 10^{+03}$	$3.75 \times 10^{+03}$	$2.46 \times 10^{+03}$	$2.18 \times 10^{+03}$
F17	std	$4.49 \times 10^{+02}$	$7.10 \times 10^{+03}$	$3.42 \times 10^{+02}$	$4.76 \times 10^{+02}$	$1.68 \times 10^{+03}$	$4.36 \times 10^{+02}$	$4.78 \times 10^{+02}$	$3.01 \times 10^{+02}$
F17	avg	$3.89 \times 10^{+03}$	$1.33 \times 10^{+04}$	$3.33 \times 10^{+03}$	$4.33 \times 10^{+03}$	$7.28 \times 10^{+03}$	$4.58 \times 10^{+03}$	$3.44 \times 10^{+03}$	$2.75 \times 10^{+03}$
F18	min	$1.44 \times 10^{+06}$	$3.35 \times 10^{+07}$	$1.14 \times 10^{+05}$	$1.55 \times 10^{+06}$	$4.70 \times 10^{+07}$	$7.54 \times 10^{+06}$	$1.07 \times 10^{+05}$	$9.17 \times 10^{+04}$
F18	std	$6.45 \times 10^{+06}$	$1.26 \times 10^{+08}$	$7.21 \times 10^{+05}$	$1.72 \times 10^{+07}$	$6.86 \times 10^{+07}$	$2.61 \times 10^{+07}$	$1.19 \times 10^{+06}$	$7.76 \times 10^{+05}$
F18	avg	$8.59 \times 10^{+06}$	$2.08 \times 10^{+08}$	$7.91 \times 10^{+05}$	$2.02 \times 10^{+07}$	$1.56 \times 10^{+08}$	$3.83 \times 10^{+07}$	$8.84 \times 10^{+05}$	$1.02 \times 10^{+06}$
F19	min	$3.76 \times 10^{+05}$	$2.31 \times 10^{+03}$	$2.11 \times 10^{+03}$	$2.02 \times 10^{+03}$	$1.80 \times 10^{+09}$	$4.02 \times 10^{+06}$	$4.17 \times 10^{+03}$	$3.46 \times 10^{+03}$
F19	std	$1.99 \times 10^{+06}$	$1.37 \times 10^{+04}$	$1.26 \times 10^{+04}$	$1.05 \times 10^{+07}$	$7.93 \times 10^{+08}$	$7.60 \times 10^{+07}$	$7.87 \times 10^{+03}$	$7.09 \times 10^{+01}$
F19	avg	$1.99 \times 10^{+06}$	$1.97 \times 10^{+04}$	$2.07 \times 10^{+04}$	$7.86 \times 10^{+06}$	$3.44 \times 10^{+09}$	$6.86 \times 10^{+07}$	$1.64 \times 10^{+04}$	$3.57 \times 10^{+03}$
F20	min	$3.00 \times 10^{+03}$	$3.62 \times 10^{+03}$	$2.67 \times 10^{+03}$	$2.97 \times 10^{+03}$	$3.82 \times 10^{+03}$	$3.86 \times 10^{+03}$	$2.53 \times 10^{+03}$	$2.21 \times 10^{+03}$
F20	std	$3.01 \times 10^{+02}$	$2.20 \times 10^{+02}$	$2.81 \times 10^{+02}$	$3.80 \times 10^{+02}$	$1.68 \times 10^{+02}$	$1.85 \times 10^{+02}$	$5.32 \times 10^{+02}$	$2.79 \times 10^{+02}$
F20	avg	$3.49 \times 10^{+03}$	$4.36 \times 10^{+03}$	$3.25 \times 10^{+03}$	$3.78 \times 10^{+03}$	$4.18 \times 10^{+03}$	$4.23 \times 10^{+03}$	$3.48 \times 10^{+03}$	$2.90 \times 10^{+03}$
F21	min	$2.77 \times 10^{+03}$	$3.12 \times 10^{+03}$	$2.58 \times 10^{+03}$	$2.73 \times 10^{+03}$	$3.10 \times 10^{+03}$	$2.82 \times 10^{+03}$	$2.47 \times 10^{+03}$	$2.41 \times 10^{+03}$
F21	std	$8.80 \times 10^{+01}$	$8.80 \times 10^{+01}$	$5.73 \times 10^{+01}$	$9.36 \times 10^{+01}$	$4.78 \times 10^{+01}$	$5.85 \times 10^{+01}$	$7.95 \times 10^{+01}$	$2.35 \times 10^{+01}$
F21	avg	$2.95 \times 10^{+03}$	$3.27 \times 10^{+03}$	$2.66 \times 10^{+03}$	$2.88 \times 10^{+03}$	$3.20 \times 10^{+03}$	$2.95 \times 10^{+03}$	$2.59 \times 10^{+03}$	$2.45 \times 10^{+03}$
F22	min	$9.91 \times 10^{+03}$	$1.50 \times 10^{+04}$	$9.09 \times 10^{+03}$	$3.96 \times 10^{+03}$	$1.63 \times 10^{+04}$	$9.26 \times 10^{+03}$	$2.39 \times 10^{+03}$	$2.34 \times 10^{+03}$
F22	std	$9.32 \times 10^{+02}$	$7.31 \times 10^{+02}$	$1.66 \times 10^{+03}$	$2.83 \times 10^{+03}$	3.73×1			

Table 3: Continued

		HHO	BOA	GTO	DBO	BWO	SABO	APO	CAAPO
F23	min	$3.60 \times 10^{+03}$	$4.22 \times 10^{+03}$	$3.03 \times 10^{+03}$	$3.21 \times 10^{+03}$	$4.01 \times 10^{+03}$	$3.63 \times 10^{+03}$	$2.96 \times 10^{+03}$	$2.85 \times 10^{+03}$
F23	std	$1.85 \times 10^{+02}$	$2.10 \times 10^{+02}$	$1.87 \times 10^{+02}$	$1.61 \times 10^{+02}$	$7.33 \times 10^{+01}$	$1.40 \times 10^{+02}$	$8.45 \times 10^{+01}$	$4.00 \times 10^{+01}$
F23	avg	$3.91 \times 10^{+03}$	$4.69 \times 10^{+03}$	$3.30 \times 10^{+03}$	$3.55 \times 10^{+03}$	$4.14 \times 10^{+03}$	$3.93 \times 10^{+03}$	$3.11 \times 10^{+03}$	$2.91 \times 10^{+03}$
F24	min	$3.96 \times 10^{+03}$	$4.87 \times 10^{+03}$	$3.19 \times 10^{+03}$	$3.42 \times 10^{+03}$	$4.28 \times 10^{+03}$	$3.67 \times 10^{+03}$	$3.10 \times 10^{+03}$	$3.03 \times 10^{+03}$
F24	std	$2.81 \times 10^{+02}$	$2.80 \times 10^{+02}$	$1.33 \times 10^{+02}$	$1.77 \times 10^{+02}$	$1.39 \times 10^{+02}$	$1.22 \times 10^{+02}$	$8.60 \times 10^{+01}$	$3.92 \times 10^{+01}$
F24	avg	$4.41 \times 10^{+03}$	$5.42 \times 10^{+03}$	$3.42 \times 10^{+03}$	$3.73 \times 10^{+03}$	$4.50 \times 10^{+03}$	$3.91 \times 10^{+03}$	$3.18 \times 10^{+03}$	$3.09 \times 10^{+03}$
F25	min	$3.37 \times 10^{+03}$	$1.37 \times 10^{+04}$	$3.11 \times 10^{+03}$	$3.10 \times 10^{+03}$	$1.28 \times 10^{+04}$	$5.16 \times 10^{+03}$	$3.12 \times 10^{+03}$	$3.08 \times 10^{+03}$
F25	std	$2.42 \times 10^{+02}$	$1.12 \times 10^{+03}$	$9.15 \times 10^{+01}$	$5.68 \times 10^{+02}$	$6.77 \times 10^{+02}$	$8.64 \times 10^{+02}$	$4.70 \times 10^{+01}$	$4.09 \times 10^{+01}$
F25	avg	$3.76 \times 10^{+03}$	$1.58 \times 10^{+04}$	$3.24 \times 10^{+03}$	$3.61 \times 10^{+03}$	$1.43 \times 10^{+04}$	$7.10 \times 10^{+03}$	$3.18 \times 10^{+03}$	$3.18 \times 10^{+03}$
F26	min	$8.28 \times 10^{+03}$	$1.62 \times 10^{+04}$	$3.82 \times 10^{+03}$	$4.58 \times 10^{+03}$	$1.59 \times 10^{+04}$	$1.01 \times 10^{+04}$	$5.63 \times 10^{+03}$	$3.36 \times 10^{+03}$
F26	std	$1.08 \times 10^{+03}$	$6.28 \times 10^{+02}$	$2.44 \times 10^{+03}$	$2.00 \times 10^{+03}$	$3.75 \times 10^{+02}$	$1.10 \times 10^{+03}$	$9.05 \times 10^{+02}$	$1.05 \times 10^{+03}$
F26	avg	$1.18 \times 10^{+04}$	$1.76 \times 10^{+04}$	$7.49 \times 10^{+03}$	$1.09 \times 10^{+04}$	$1.69 \times 10^{+04}$	$1.35 \times 10^{+04}$	$7.55 \times 10^{+03}$	$4.08 \times 10^{+03}$
F27	min	$4.23 \times 10^{+03}$	$5.21 \times 10^{+03}$	$3.52 \times 10^{+03}$	$3.45 \times 10^{+03}$	$5.46 \times 10^{+03}$	$4.00 \times 10^{+03}$	$3.52 \times 10^{+03}$	$3.39 \times 10^{+03}$
F27	std	$5.75 \times 10^{+02}$	$7.98 \times 10^{+02}$	$3.35 \times 10^{+02}$	$2.48 \times 10^{+02}$	$4.15 \times 10^{+02}$	$4.37 \times 10^{+02}$	$8.39 \times 10^{+01}$	$7.34 \times 10^{+01}$
F27	avg	$5.09 \times 10^{+03}$	$6.94 \times 10^{+03}$	$3.97 \times 10^{+03}$	$3.96 \times 10^{+03}$	$6.24 \times 10^{+03}$	$4.54 \times 10^{+03}$	$3.65 \times 10^{+03}$	$3.53 \times 10^{+03}$
F28	min	$4.28 \times 10^{+03}$	$1.15 \times 10^{+04}$	$3.44 \times 10^{+03}$	$3.64 \times 10^{+03}$	$1.02 \times 10^{+04}$	$5.80 \times 10^{+03}$	$3.42 \times 10^{+03}$	$3.41 \times 10^{+03}$
F28	std	$3.76 \times 10^{+02}$	$1.40 \times 10^{+03}$	$1.39 \times 10^{+02}$	$2.05 \times 10^{+03}$	$6.32 \times 10^{+02}$	$8.26 \times 10^{+02}$	$5.63 \times 10^{+01}$	$7.75 \times 10^{+01}$
F28	avg	$4.82 \times 10^{+03}$	$1.42 \times 10^{+04}$	$3.61 \times 10^{+03}$	$6.34 \times 10^{+03}$	$1.22 \times 10^{+04}$	$7.47 \times 10^{+03}$	$3.51 \times 10^{+03}$	$3.54 \times 10^{+03}$
F29	min	$5.76 \times 10^{+03}$	$3.09 \times 10^{+04}$	$4.54 \times 10^{+03}$	$4.97 \times 10^{+03}$	$1.23 \times 10^{+04}$	$7.68 \times 10^{+03}$	$3.90 \times 10^{+03}$	$3.52 \times 10^{+03}$
F29	std	$8.14 \times 10^{+02}$	$2.14 \times 10^{+05}$	$6.15 \times 10^{+02}$	$7.53 \times 10^{+02}$	$1.58 \times 10^{+04}$	$2.45 \times 10^{+03}$	$4.50 \times 10^{+02}$	$2.15 \times 10^{+02}$
F29	avg	$6.82 \times 10^{+03}$	$2.81 \times 10^{+05}$	$5.38 \times 10^{+03}$	$6.17 \times 10^{+03}$	$3.22 \times 10^{+04}$	$1.13 \times 10^{+04}$	$4.61 \times 10^{+03}$	$3.97 \times 10^{+03}$
F30	min	$4.52 \times 10^{+07}$	$3.77 \times 10^{+09}$	$7.68 \times 10^{+05}$	$2.59 \times 10^{+06}$	$2.43 \times 10^{+09}$	$1.50 \times 10^{+08}$	$2.31 \times 10^{+06}$	$1.44 \times 10^{+06}$
F30	std	$4.72 \times 10^{+07}$	$3.17 \times 10^{+09}$	$1.28 \times 10^{+06}$	$4.72 \times 10^{+07}$	$1.33 \times 10^{+09}$	$2.83 \times 10^{+08}$	$1.50 \times 10^{+06}$	$4.46 \times 10^{+05}$
F30	avg	$1.18 \times 10^{+08}$	$7.96 \times 10^{+09}$	$2.18 \times 10^{+06}$	$5.06 \times 10^{+07}$	$4.74 \times 10^{+09}$	$6.14 \times 10^{+08}$	$4.58 \times 10^{+06}$	$2.02 \times 10^{+06}$

Algorithm 1: Pseudo-Code of Proposed CAAPO.

```

Start CAAPO.
1. Input:  $N, T, D, F$  and  $C$ 
2. Initialize  $N$  puffins,  $\vec{X}_i^t (i = 1, 2, \dots, N)$ , using Equation 23
3. Evaluate each  $\vec{X}_i^t$ , and find the one with the best fitness in  $\vec{X}_i^t$ 
4.  $t = 1$ 
5. while  $t < T$ 
6. Calculate the behaviour conversion factor  $B$  according to Equation 21
7. Update  $\omega$  value using Equation 30
8. Update  $\vec{X}_i^t$  using Equation 25
9. if  $B > C$  %% Air flight period Exploration phase
10.  for  $i = 1:N$ 
11.    Update  $\vec{Y}_i^{t+1}$  using Equation 6%% Air search
12.    Update  $\vec{Z}_i^{t+1}$  using Equation 9%% Dive predation
13.    Select  $N$  excellent populations as the new population  $\vec{X}_i^{t+1}$  using Equation 11–13
14.    Evaluate the puffins,  $\vec{X}_i^{t+1}$ , and replace  $\vec{X}_i^t$  with  $\vec{X}_i^{t+1}$  if it is better.
15.     $t = t + 1$ 
16.  end for
17. else %% Diving and foraging period Exploitation phase
18.  for  $i = 1:N$ 
19.    Update  $\vec{W}_i^{t+1}$  using Equation 28%% Gather for food
20.    Update  $\vec{Y}_i^{t+1}$  using Equation 15%% Strengthen search
21.    Update  $\vec{Z}_i^{t+1}$  using Equation 29%% Avoid predators
22.    Select  $N$  excellent populations as the new population  $\vec{X}_i^{t+1}$  using Equation 18–20
23.    Evaluate the puffins,  $\vec{X}_i^{t+1}$ , and replace  $\vec{X}_i^t$  with  $\vec{X}_i^{t+1}$  if it is better.
24.     $t = t + 1$ 
25.  end for
26. end if
27. end while
28. return  $\vec{X}_i^t$ 
29. Output: the best  $\vec{X}_i^t$  and its fitness value
End CAAPO.

```

including chaotic mapping and adaptive inertia weights maintain polynomial-time complexity equivalence with the baseline APO. The lens imaging reverse learning mechanism introduces additional computational overhead during exploration phases while preserving the original asymptotic complexity. Consequently, the enhanced CAAPO variant maintains the same time complexity $O(N \cdot D \cdot T)$ as the fundamental algorithm.

5. Simulation Experiment Analysis

5.1. Experiment and environmental set-up

To validate the enhanced APO algorithm's efficacy in 3D wireless sensor network (3D-WSN) coverage optimization, this study employs standardized benchmark functions and conduct a comparative analysis against swarm intelligence metaheuristics. The experimental framework incorporates CAAPO alongside seven established metaheuristics: Harris Hawks Optimization (HHO; Heidari et al., 2019), Butterfly Optimization Algorithm (BOA; Arora & Singh, 2019), Dung Beetle Optimizer (DBO; Xue & Shen, 2023), Gorilla Troops Optimizer (GTO; Abdollahzadeh et al., 2021), Beluga Whale Optimization (BWO; Zhong et al., 2022), Subtraction-Average-Based Optimizer (SABO; Trojovský & Dehghani, 2023), and the baseline Arctic Puffin Optimization (APO). Rigorous benchmarking conducted on the CEC2017 test suite and two engineering applications evaluates CAAPO's performance superiority through tripartite metrics: optimal values, arithmetic means, and solution stability indices derived from functional optimization outcomes.

The experimental protocol standardized comparative parameters across all algorithms: 30 independent trials per method with maximum iterations $T = 500$ and population size $N = 30$. The benchmark algorithms maintained parameter configurations according to their original literature specifications. The evaluation framework incorporated three CEC2017 benchmark

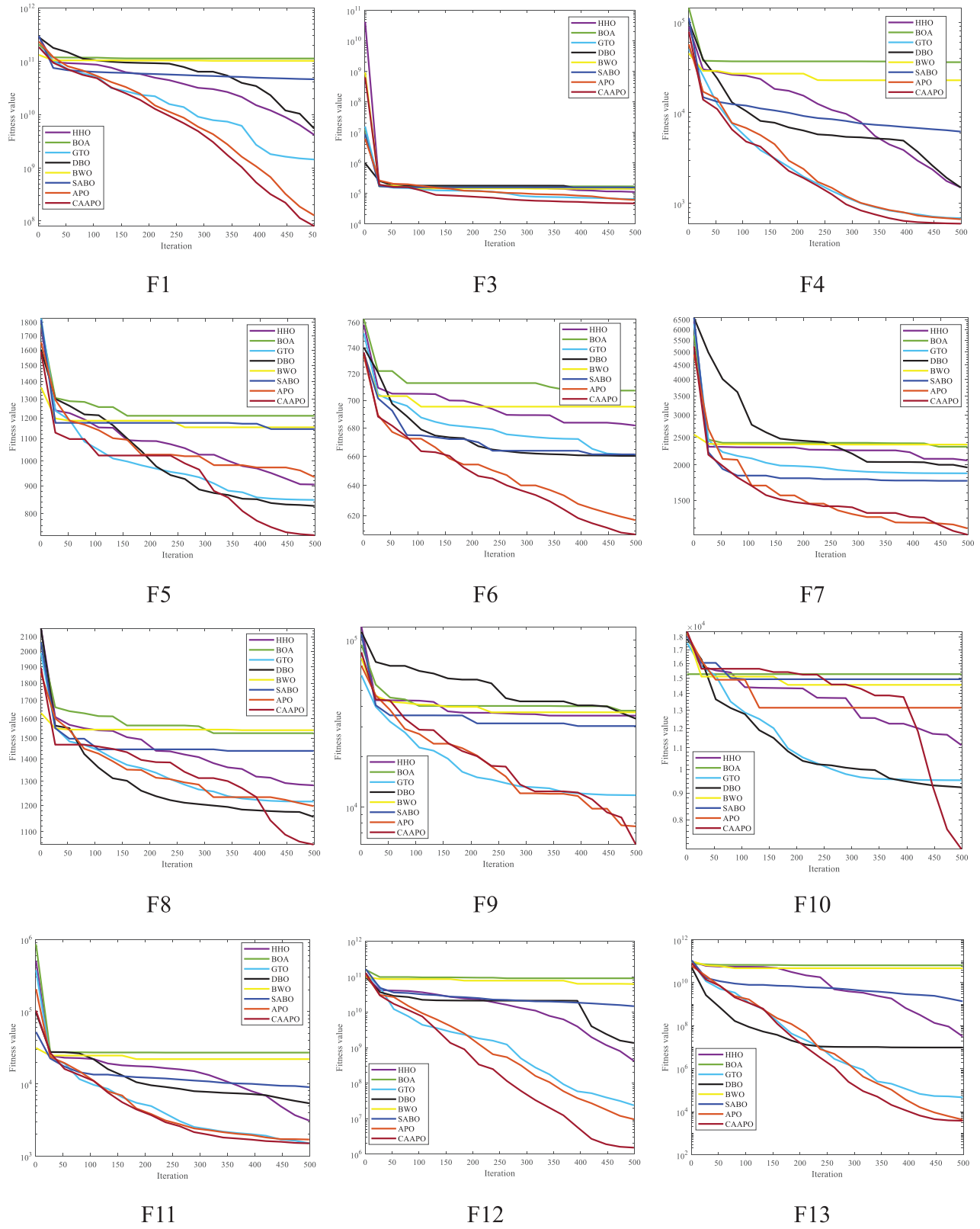
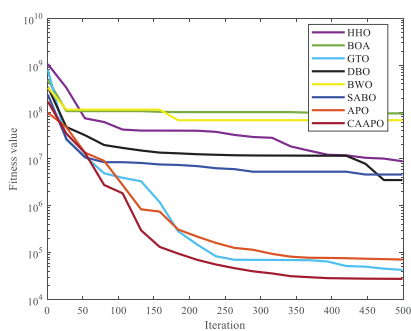
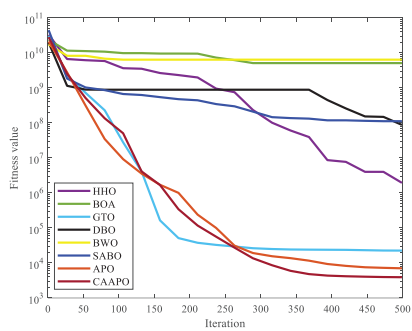


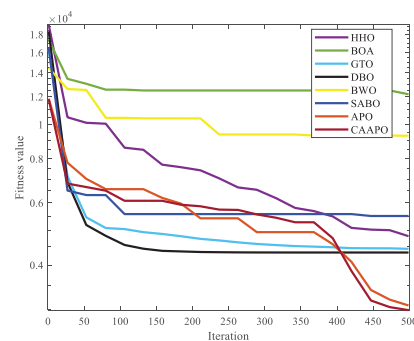
Figure 6: Iteration profile of different improved algorithms($D = 50$).



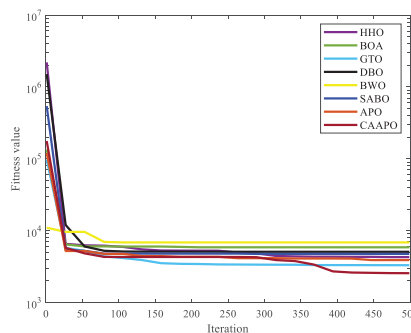
F14



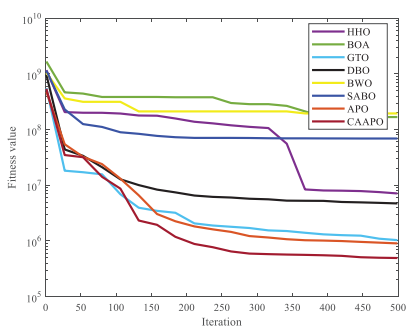
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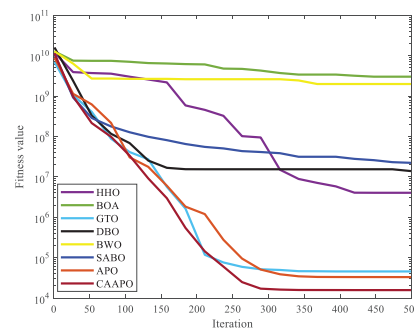
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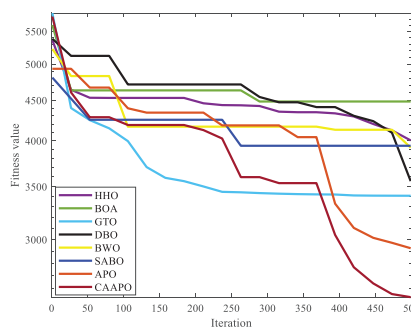
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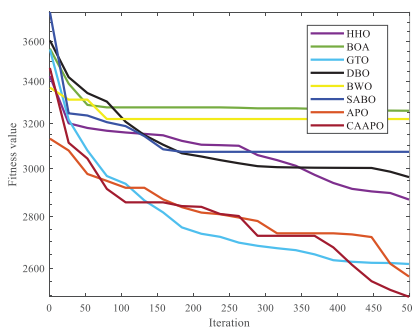
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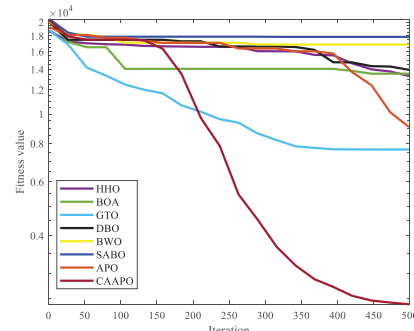
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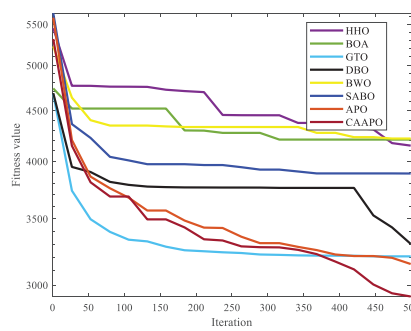
F20



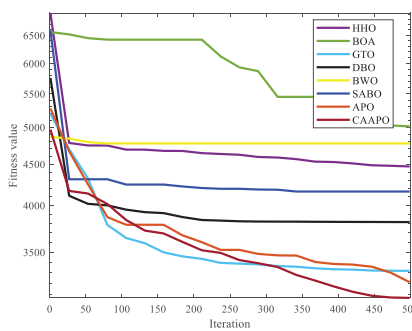
F21



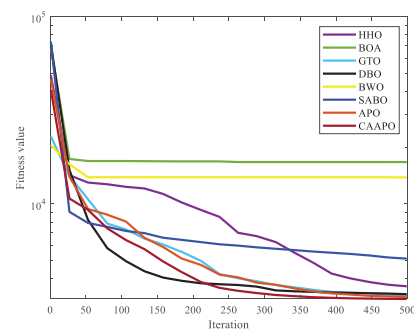
F22



F23



F24



F25

Figure 6: Continued.

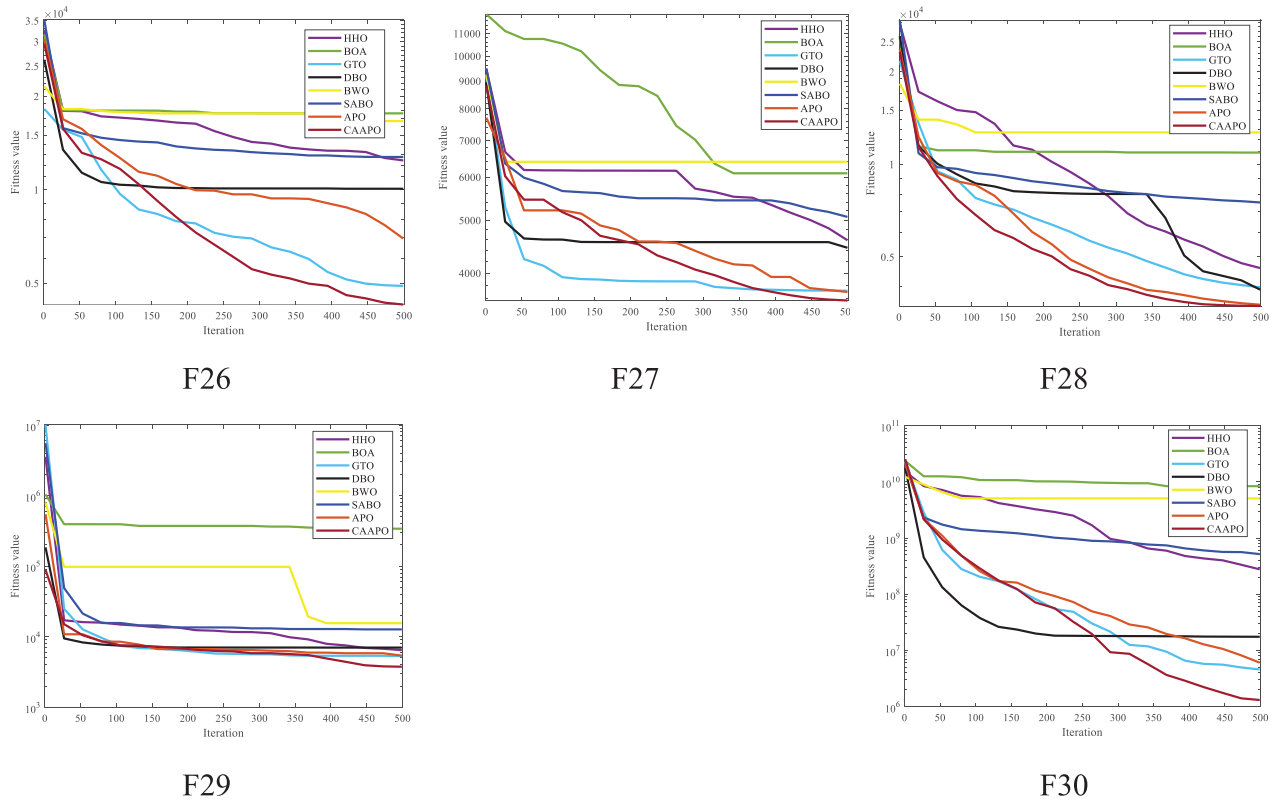


Figure 6: Continued.

categories: unimodal (exploitation assessment), multimodal (exploration evaluation), and fixed-dimensional multimodal (balanced analysis) functions. Mathematical formulations and theoretical optima for these functions are detailed in Table 1. However, due to numerical instability in MATLAB implementations, F2 was excluded from the 30-function benchmark suite. All computations were executed on MATLAB R2020b under Windows 11, utilizing an Intel Core i5-11400H processor (2.70 GHz) to ensure computational consistency.

5.2. Comparison and analysis of CAAPO with other metaheuristic algorithms

The CAAPO algorithm's optimization capability was evaluated against state-of-the-art metaheuristics using the CEC2017 benchmark suite (dimensionality $D = 30$). Quantitative results from 30 independent trials are presented in Table 2, documenting global minima attainment (min), arithmetic mean fitness (avg), and solution stability metric (std). Visual analytics reveal convergence characteristics through dual modalities: Figure 4 illustrates evolutionary trajectory dynamics across iterations, while Figure 5 presents statistical distribution analysis using boxplot visualization.

Table 2 demonstrates CAAPO's capability to achieve global optima across all test functions except F1, F10, and F28, with triple-criterion dominance (minimum, mean, standard deviation) in 14 instances. This performance profile confirms the enhanced Arctic puffin methodology's superior optimization precision. In unimodal functions (F1, F3) and basic multimodal functions (F4–F10), CAAPO marginally underperforms GTO and DBO in optimization precision. Although suboptimal in F3, F4, and F9 metrics, CAAPO maintains statistical parity with leading solutions. The algorithm

exhibits enhanced optimization efficacy in hybrid (F11–F20) and composite functions (F21–F30), demonstrating superior accuracy-efficiency trade-off. Notably, CAAPO achieves maximum optimal values in mixed-function scenarios with exceptional solution fidelity. The methodology demonstrates triple-criterion dominance in seven functions, with notable variance reduction in F25 and F28 compared to those of baseline APO while maintaining comparable mean values, substantiating its enhanced optimization stability.

While the enhanced CAAPO algorithm exhibits residual susceptibility to local optima in specific function classes, Figure 4 illustrates its improved escape dynamics through adaptive inertia weighting and lens imaging inversion mechanisms during late-stage iterations. Empirical validation reveals CAAPO's superior convergence fidelity compared to baseline APO in functions F20, F21, and F26, where traditional methodology demonstrates persistent local convergence. Quantile analysis of solution distributions confirms CAAPO's enhanced stability, evidenced by reduced outlier frequency and lower variance metric across trials. The algorithm particularly excels in complex function optimization, maintaining upper quartile dominance in high-dimensional test cases. Comparative analysis substantiates CAAPO's superior convergence efficiency and operational robustness relative to conventional approaches.

5.3. Further comparative experiments of the algorithm

To validate the enhanced algorithm's efficacy in high-dimensional optimization, this study evaluates eight metaheuristics on the CEC2017 benchmark suite (dimensionality $D = 50$). Quantitative results in Table 3 document arithmetic means, standard deviations, and global optima attainment, with optimal outcomes

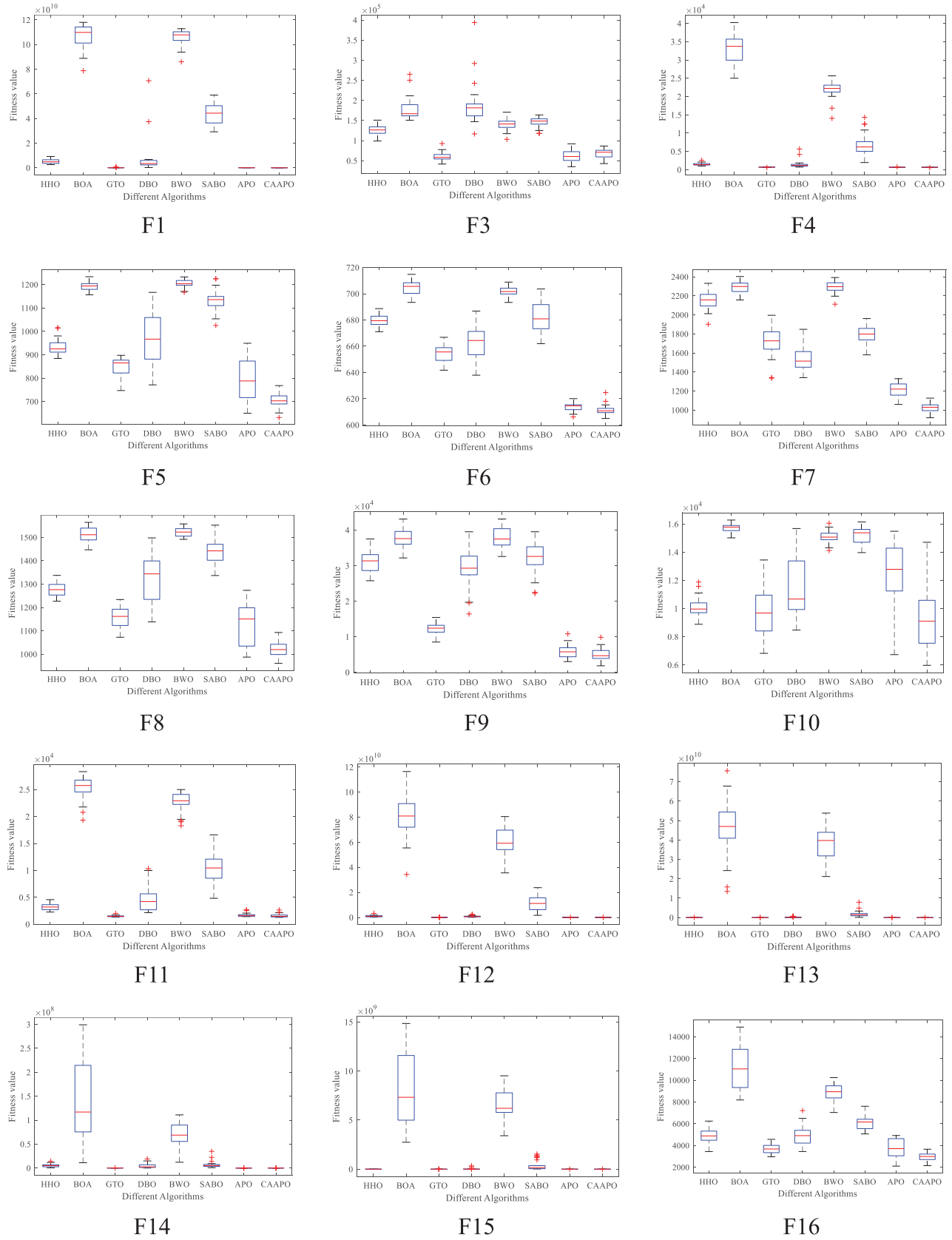


Figure 7: Box plots of iterative data distributions($D = 50$).

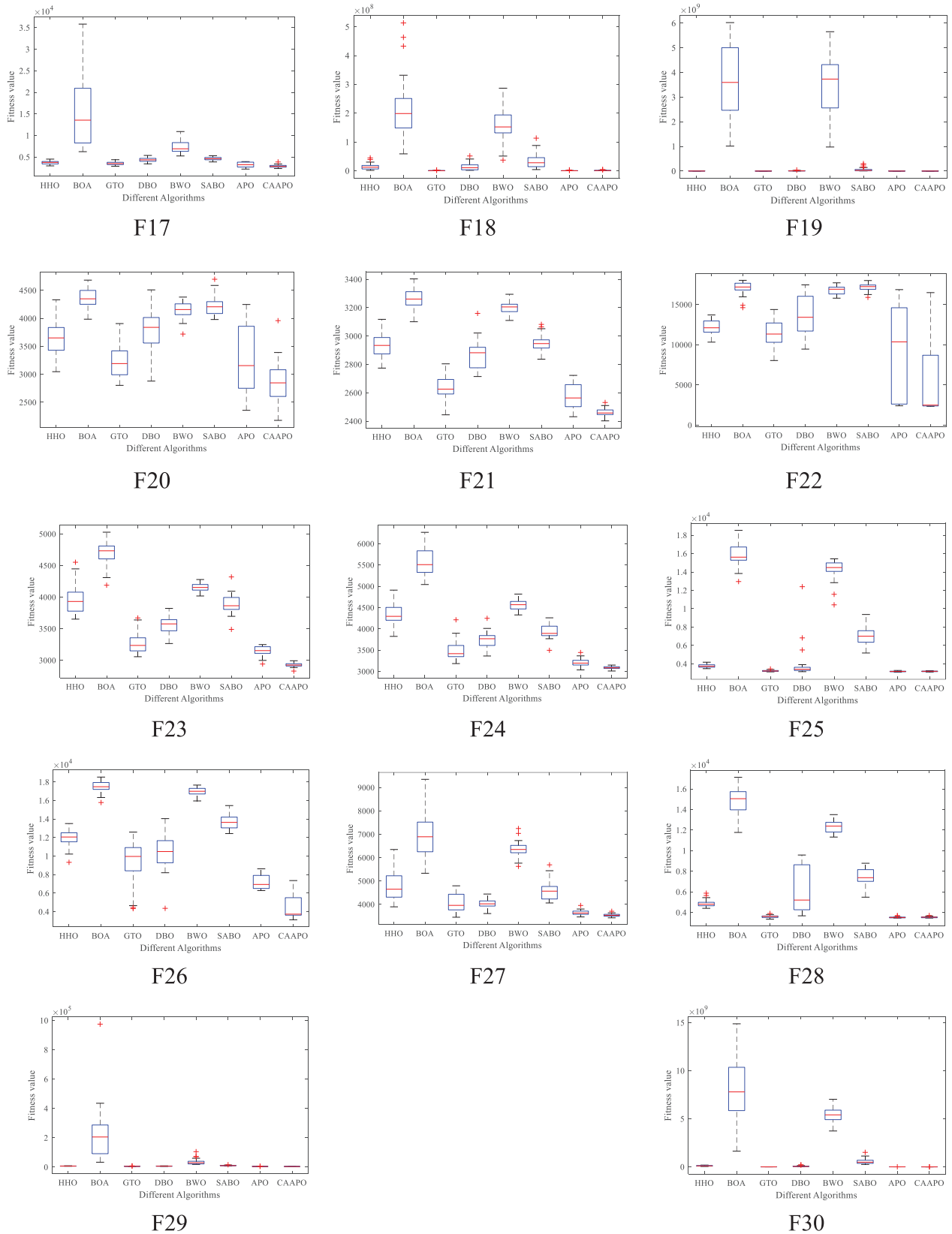


Figure 7: Continued.

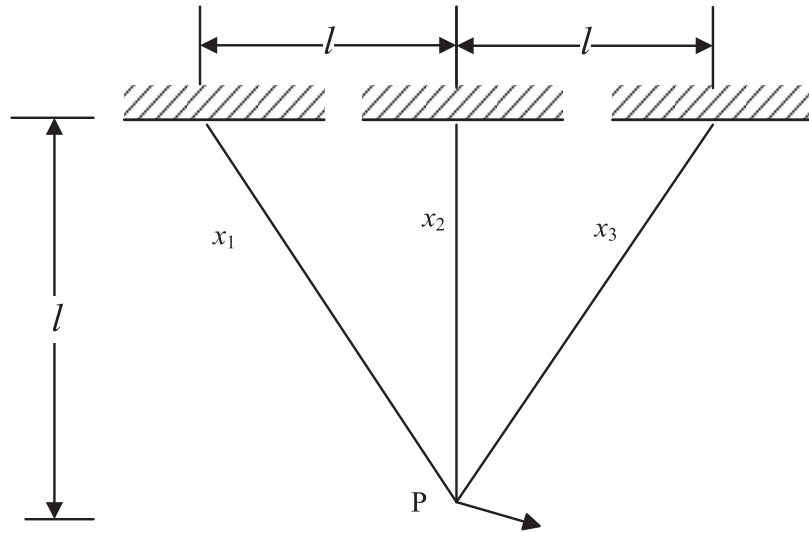


Figure 8: Schematic diagram of the design optimization problem for a Three-Rod Truss.

Table 4: Results of solving the optimization problem for three-bar truss design.

Name	x_1	x_3	Best	Std	Mean
CAAPO	7.65×10^{-01}	3.96×10^{-01}	$2.60 \times 10^{+02}$	4.03×10^{-07}	$2.60 \times 10^{+02}$
APO	7.65×10^{-01}	3.96×10^{-01}	$2.60 \times 10^{+02}$	1.96×10^{-06}	$2.60 \times 10^{+02}$
HHO	7.62×10^{-01}	4.04×10^{-01}	$2.60 \times 10^{+02}$	1.28×10^{-01}	$2.60 \times 10^{+02}$
BOA	7.68×10^{-01}	3.89×10^{-01}	$2.60 \times 10^{+02}$	2.18×10^{-01}	$2.60 \times 10^{+02}$
GTO	7.65×10^{-01}	3.96×10^{-01}	$2.60 \times 10^{+02}$	3.79×10^{-14}	$2.60 \times 10^{+02}$
DBO	7.65×10^{-01}	3.96×10^{-01}	$2.60 \times 10^{+02}$	2.02×10^{-13}	$2.60 \times 10^{+02}$
BWO	7.62×10^{-01}	4.05×10^{-01}	$2.60 \times 10^{+02}$	2.82×10^{-01}	$2.60 \times 10^{+02}$
SABO	7.65×10^{-01}	3.97×10^{-01}	$2.60 \times 10^{+02}$	2.84×10^{-01}	$2.60 \times 10^{+02}$

emphasized in boldface. Convergence characteristics are visualized through dual modalities: Figure 6 illustrates evolutionary trajectory dynamics; Figure 7 presents solution distribution analysis via boxplot visualization. These comparative analyses substantiate the methodology's capability to maintain solution fidelity while addressing high-dimensional complexity constraints.

The experimental analysis in Table 3 demonstrates the CAAPO algorithm's superior optimization capability, achieving global minima in 27/29 test functions and triple-criterion dominance (minimum, mean, standard deviation) across 15 instances. This performance profile confirms the methodology's enhancement efficacy. While exhibiting marginally inferior standard deviation in specific functions (F4, F14, F28) compared to baseline APO, CAAPO maintains superior mean values and solution quality, indicating broader applicability. Notably, the algorithm maintains consistent efficacy in 50-dimensional multimodal and composite function optimization, successfully overcoming local optima constraints to attain global solutions – a capability visually substantiated by the convergence dynamics in Figure 6.

The F20 case study reveals that both CAAPO and APO exhibit local convergence trapping at 50, 120, and 200 iterations. However, CAAPO's lens imaging inversion mechanism significantly enhances local optima convergence dynamics, achieving global solution escape post-230 iterations. The methodology demonstrates rapid solution space re-exploration capability, reacquiring global optima within 20 subsequent iterations and maintaining persistent non-regressive convergence beyond 350 iterations. This en-

hanced performance stems from adaptive convergence operators that balance exploration-exploitation trade-off during late-stage optimization. Quantitative validation through Figure 7's boxplot analysis confirms CAAPO's statistical superiority via compact interquartile ranges and outlier suppression, substantiating the methodology's improved robustness in high-dimensional landscapes.

5.4. Engineering design issues

This study implements the enhanced Arctic Puffin Optimization (CAAPO) algorithm in engineering applications, conducting simulation experiments with comparative analysis against established optimization methodologies to validate practical effectiveness and operational viability.

5.4.1. Three-bar truss design optimization problem

The three-rod truss represents a fundamental structural configuration in engineering mechanics, serving as critical load-bearing component in large-scale infrastructure and mechanical systems. This optimization problem has two continuous design variables (x_1, x_2) that represent cross-sectional areas under symmetry constraint ($x_1 = x_3$). The objective is to minimize total structural volume through geometric parameter optimization, as illustrated in Figure 8. Each member is governed by stress constraints, formulating a non-linear constrained optimization problem with three inequality constraints.

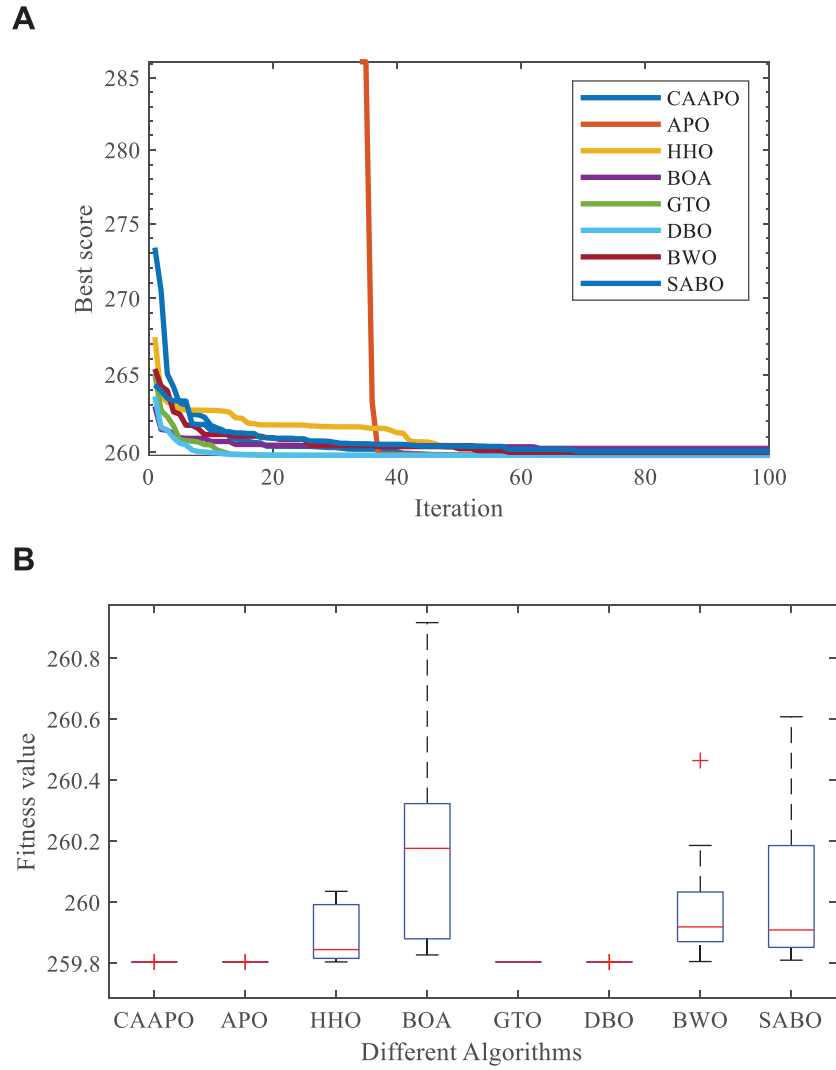


Figure 9: Iterative profile and box plots of the three-bar truss design optimization problem. (A) Iteration curve graph (B) Iterative data distribution box plot.

Table 5: Gear Train Design Problem Solving Results.

name	x_1	x_2	x_3	x_4	best	std	mean
CAAPO	$5.34 \times 10^{+01}$	$1.32 \times 10^{+01}$	$1.96 \times 10^{+01}$	$3.38 \times 10^{+01}$	2.31×10^{-11}	9.08×10^{-10}	1.23×10^{-09}
APO	$4.94 \times 10^{+01}$	$1.26 \times 10^{+01}$	$3.06 \times 10^{+01}$	$5.68 \times 10^{+01}$	9.94×10^{-11}	9.14×10^{-09}	7.06×10^{-09}
HHO	$5.75 \times 10^{+01}$	$1.20 \times 10^{+01}$	$3.75 \times 10^{+01}$	$5.37 \times 10^{+01}$	8.89×10^{-10}	8.08×10^{-09}	5.10×10^{-09}
BOA	$4.78 \times 10^{+01}$	$3.04 \times 10^{+01}$	$1.22 \times 10^{+01}$	$5.18 \times 10^{+01}$	2.36×10^{-09}	2.41×10^{-03}	1.73×10^{-03}
GTO	$5.07 \times 10^{+01}$	$2.56 \times 10^{+01}$	$1.50 \times 10^{+01}$	$5.32 \times 10^{+01}$	2.31×10^{-11}	1.28×10^{-08}	8.81×10^{-09}
DBO	$4.55 \times 10^{+01}$	$1.30 \times 10^{+01}$	$2.37 \times 10^{+01}$	$4.74 \times 10^{+01}$	9.92×10^{-10}	2.35×10^{-08}	2.20×10^{-08}
BWO	$3.34 \times 10^{+01}$	$1.73 \times 10^{+01}$	$1.45 \times 10^{+01}$	$4.97 \times 10^{+01}$	1.36×10^{-09}	2.26×10^{-08}	2.32×10^{-08}
SABO	$5.73 \times 10^{+01}$	$1.28 \times 10^{+01}$	$3.14 \times 10^{+01}$	$4.88 \times 10^{+01}$	9.94×10^{-11}	1.76×10^{-09}	2.12×10^{-09}

Objective function:

$$\text{Variable } x = [x_1, x_2]; \quad (31)$$

$$\min f(x) = (2\sqrt{2} \cdot x_1 + x_2) \cdot l. \quad (32)$$

Constraints:

$$g_1(x) = \frac{\sqrt{2} \cdot x_1 + x_2}{\sqrt{2} \cdot x_1^2 + 2x_1x_2} P - \sigma \leq 0; \quad (33)$$

Value range:

$$g_2(x) = \frac{x_2}{\sqrt{2} \cdot x_1^2 + 2x_1x_2} P - \sigma \leq 0; \quad (34)$$

$$g_3(x) = \frac{1}{\sqrt{2} \cdot x_2 + x_1} P - \sigma \leq 0. \quad (35)$$

$$0 \leq x_i \leq 1, i = 1, 2; \quad (36)$$

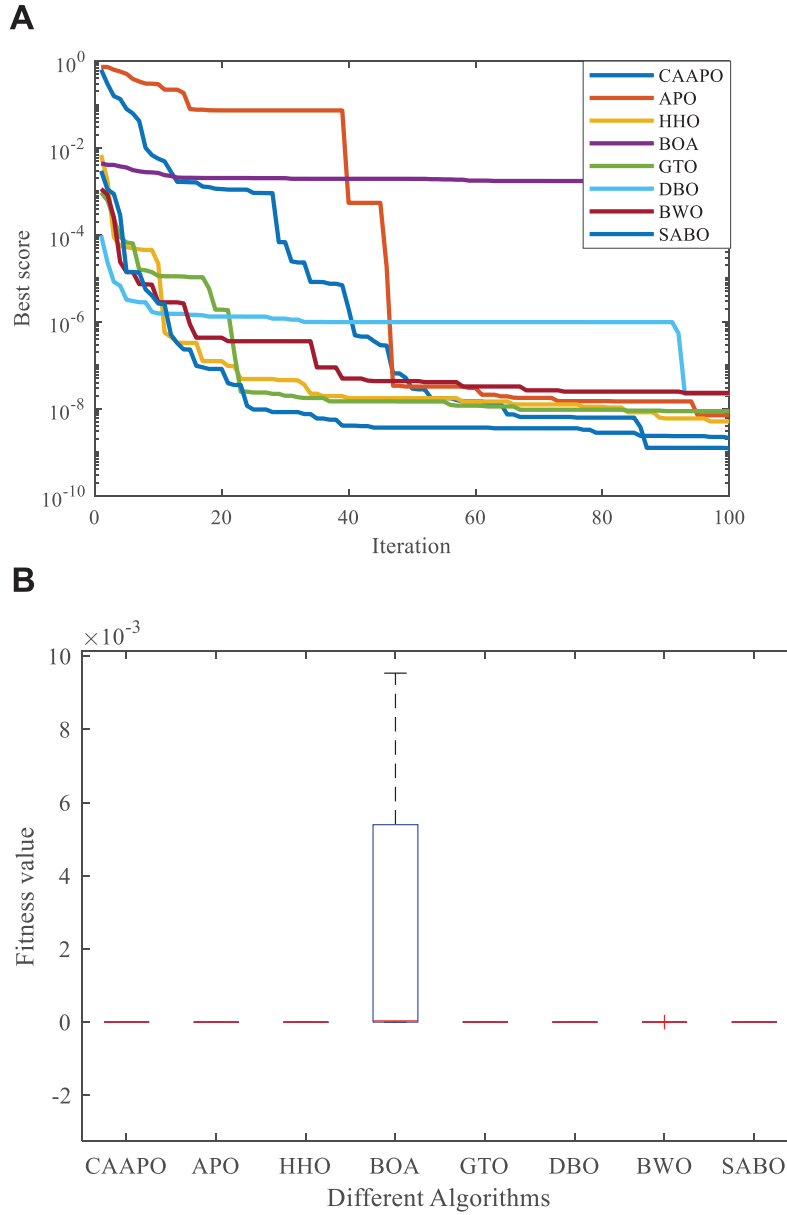


Figure 10: Iterative profile and box plots for Gear train design problem. (A) Iteration curve graph. (B) Iterative data distribution box plot.

$$l = 100 \text{ cm}; \quad (37)$$

$$P = 2 \text{ kN/cm}^2; \quad (38)$$

$$\delta = 2 \text{ kN/cm}^2. \quad (39)$$

This study evaluates the enhanced CAAPO algorithm against seven metaheuristics (APO, HHO, BOA, GTO, DBO, BWO, SABO) in structural optimization scenarios. The study standardized experimental parameters: population size $N = 30$, maximum iterations $T = 100$, with 10 independent trials per configuration. The higher performance of CAAPO in achieving global minima and solution stability is demonstrated by quantitative analysis of the three-bar truss optimization (Table 4). The improved starting population quality of CAAPO is contrasted with the inadequate early convergence of APO, as seen by the visual study of convergence dynamics (Figure 9). The box plot also shows that the modified method has higher quartiles, indicating that it performs

well in actual engineering applications and has some practical usefulness.

5.4.2. Gear train design

The optimization objective of the gear design problem is to minimize the cost of the gear ratio by reasonably designing the number of teeth for the four types of gears, which is usually described as an optimization model with inequality constraints. As shown in Equation 40. Among them, x_1 , x_2 , x_3 , and x_4 represent the number of teeth of the four different gears, respectively:

$$\min f(x) = \left(\frac{1}{6.931} - \frac{x_3 \cdot x_2}{x_1 \cdot x_4} \right)^2. \quad (40)$$

The range of values is

$$\begin{aligned} 12 &\leq x_1 \leq 60, \\ 12 &\leq x_2 \leq 60, \\ 12 &\leq x_3 \leq 60, \\ 12 &\leq x_4 \leq 60. \end{aligned} \quad (41)$$

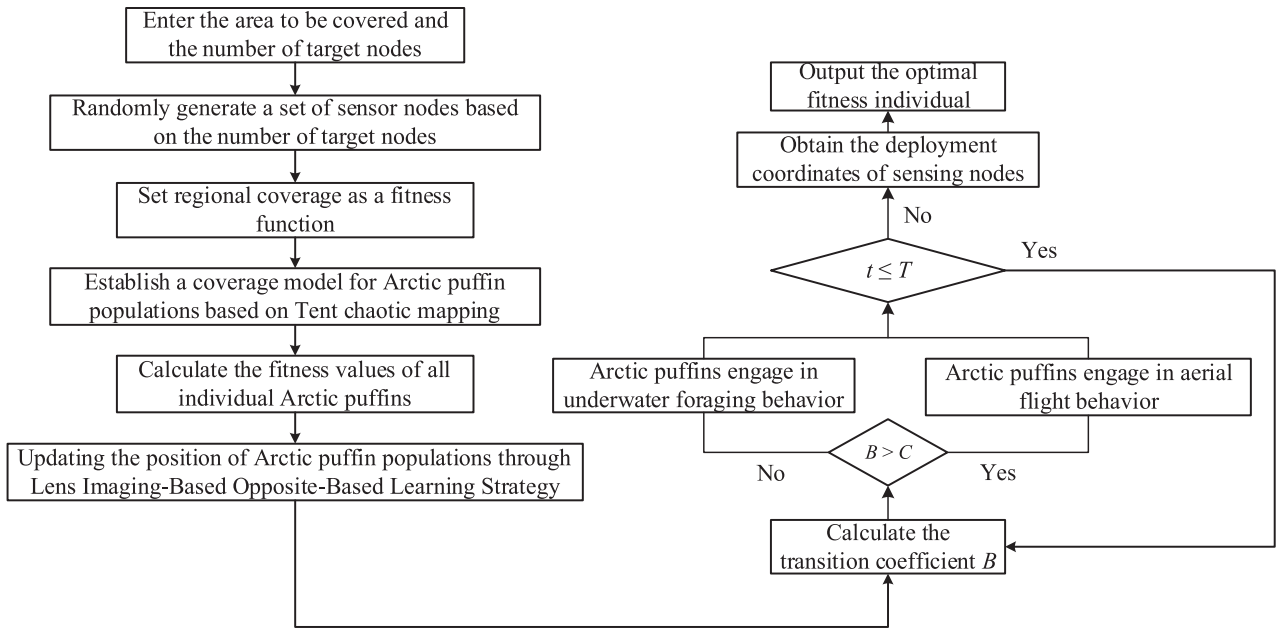


Figure 11: WSN coverage optimization flowchart.

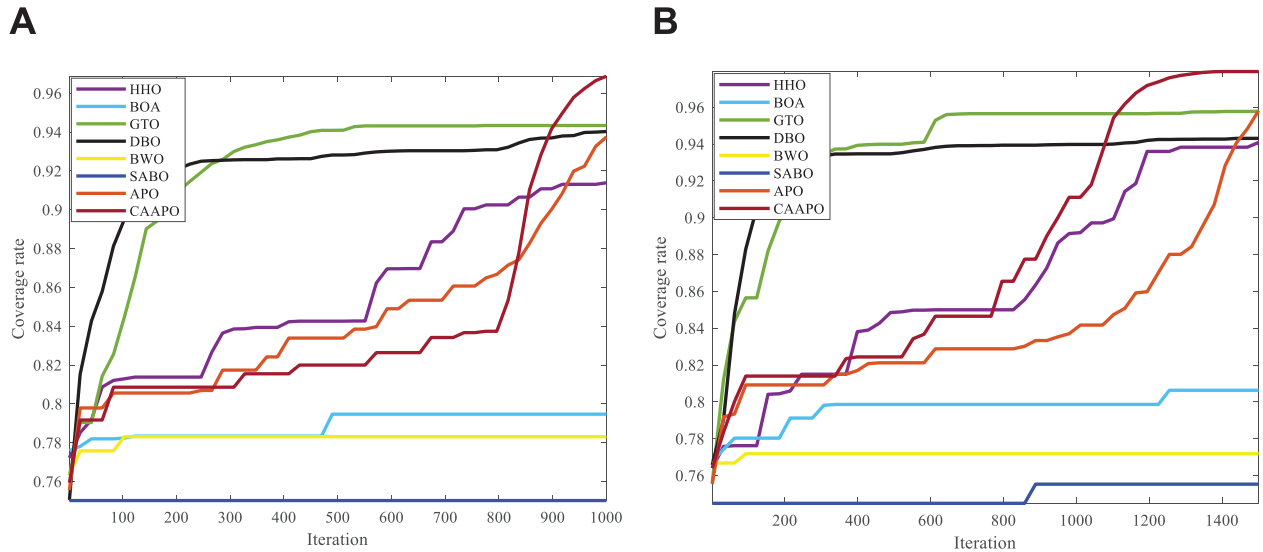


Figure 12: Algorithm comparison iteration chart. (A) $T = 1000$ (B) $T = 1500$.

The gear system design problem's optimization results are shown in Table 5, and Figure 10 uses evolutionary trajectories and statistical distribution analysis to visualize convergence dynamics. Quantitative results demonstrate CAAPO's superior solution quality, attaining both global minima and enhanced solution stability. Convergence analysis reveals critical behavioural divergence: the baseline APO algorithm exhibits persistent local convergence entrapment during early iterations (cycles 1–50), significantly impeding initial search efficiency. In contrast, CAAPO's dual-phase optimization algorithm demonstrates enhanced local optima escape capability through non-linear trajectory modulation. This empirical validation substantiates the efficacy of the tri-strategy enhancement paradigm in engineering applications,

confirming the modified Arctic Puffin methodology's capacity to address real-world mechanical design challenges with improved convergence robustness.

6. WSN Coverage Optimization Simulation Experiment and Analysis

Through simulated studies using MATLAB 2020b, the effectiveness of CAAPO in enhancing the coverage of WSNs was experimentally validated. By comparing 2D/3D nodal deployment strategies with well-known metaheuristics, this two-phase experimental approach assesses the effectiveness of geographic coverage as well as the capacity for 3D topology optimization.

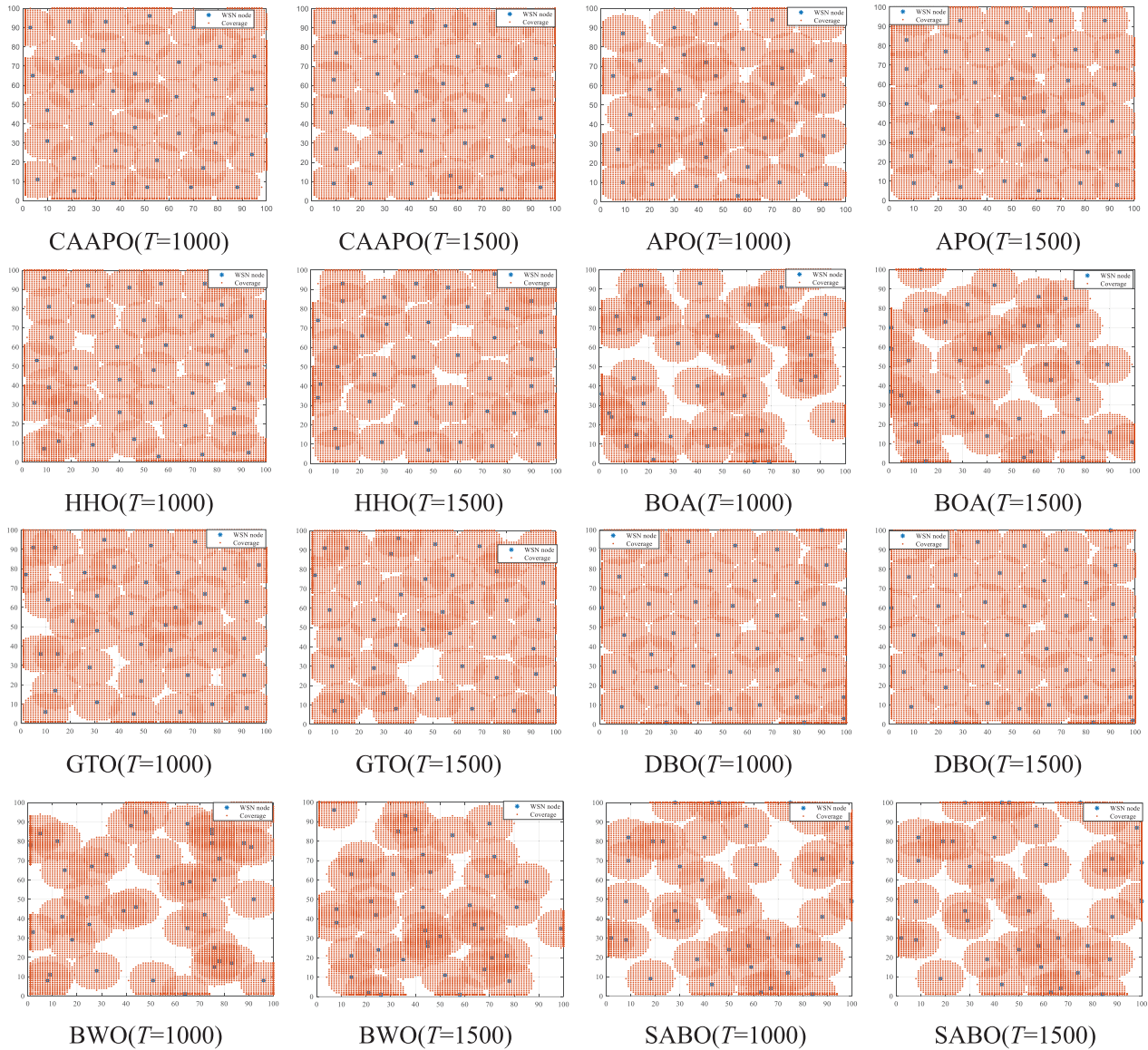


Figure 13: WSN coverage effectiveness diagram.

6.1. WSN coverage optimization process

The methodology initiates by establishing the target area's geometric boundaries and nodal coordinate distribution. A coverage fitness function is formulated through stochastic sensor node selection, followed by coordinate optimization using the Arctic Puffin Swarm Algorithm with chaotic initialization strategies to enhance spatial search precision. Figure 11 shows a schematic representation of this approach.

6.2. Deployment of 2D sensor nodes

Establish the following experimental parameters: the sensing radius ($r = 10$), the number of nodes ($N = 40$), and the monitoring area (100×100). This scenario is more in line with real-world network conditions and falls under the category of WSN deployment over a vast region. A population size of 50 people and maximum iteration counts T of 1000 and 1500, respectively, are used to establish the algorithm parameters. While Figure 12 shows the iteration curves after the CAAPO algorithm, APO algorithm, and six more optimization algorithms are executed, Figure 13 shows the

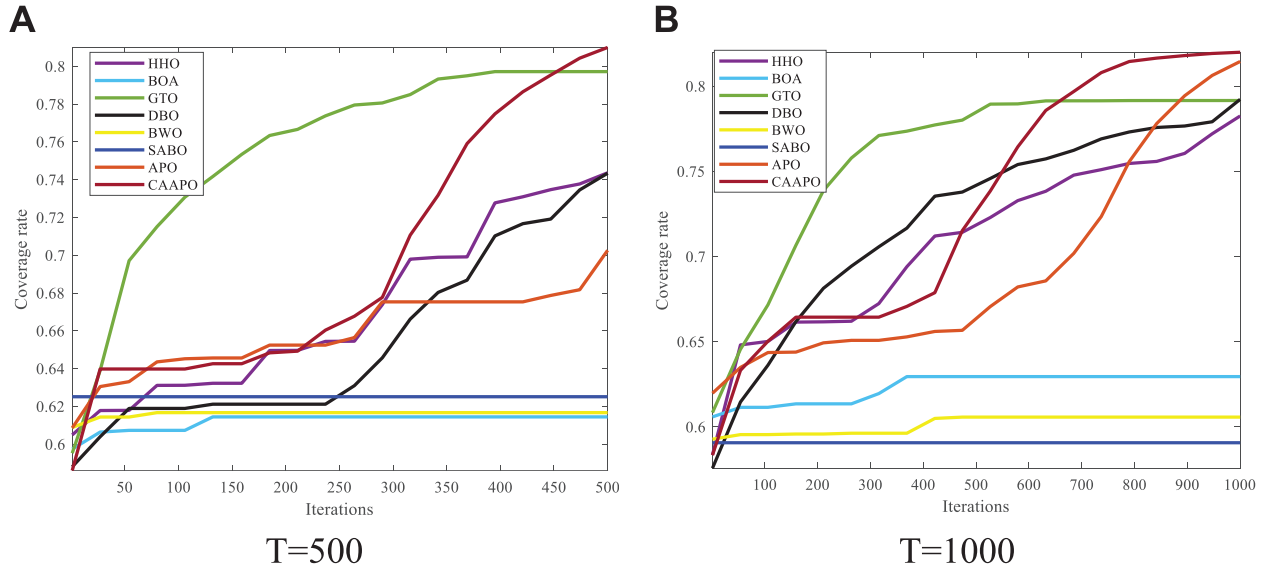
influence on node coverage after algorithm optimization. Table 6 displays the eight algorithms' running outcomes.

Figure 12 demonstrates distinct convergence characteristics across evaluated algorithms. The BOA, BWO, and SABO algorithms exhibit premature convergence due to persistent local optima entrapment throughout iterations. While DBO and GTO achieve accelerated initial convergence, both algorithms encounter significant local convergence stagnation at iterations 200 and 400, respectively, failing to achieve global optima.

The baseline APO algorithm prioritizes global search through transition coefficient B and parameter C modulation, resulting in deliberate initial-phase convergence that subsequently accelerates post-300 iterations. In contrast, the enhanced CAAPO methodology implements adaptive inertia weighting: initially moderating convergence speed to prevent premature exploitation, then dynamically intensifying search precision through phase-dependent inertia adjustment. This mechanism enables CAAPO to achieve global optima within 200 iterations post-800 cycles. It achieves faster terminal convergence than the original algorithm. Extended experimentation with $T = 1500$

Table 6: Running results of 2D sensor nodes.

T	CAAPO	APO	HHO	BOA	GTO	DBO	BWO	SABO
1000	96.88%	93.75%	91.39%	79.47%	94.34%	94.03%	78.31%	75.02%
1500	97.94%	95.82%	94.09%	80.63%	95.78%	94.31%	77.18%	75.53%

**Figure 14:** Algorithm Comparison Iteration Chart. (A) $T = 500$. (B) $T = 1000$.**Table 7:** Running results of 3D sensor nodes.

T	CAAPO	APO	HHO	BOA	GTO	DBO	BWO	SABO
500	81.00%	70.26%	74.36%	61.44%	79.72%	74.34%	61.67%	62.51%
1000	83.05%	82.80%	76.45%	61.62%	80.36%	77.82%	62.73%	59.49%

iterations confirms the dual-phase efficacy of CAAPO's inertia adaptation: enhancing global exploration capacity during initial stages while sustaining convergence velocity during deep exploitation phases, as evidenced by maintained solution stability.

Comparative analysis of coverage optimization efficacy (Table 6) reveals CAAPO achieves 3.13% and 2.12% coverage improvement over APO at 1000 and 1500 iterations respectively. Visual validation through nodal deployment patterns (Figures 12, 13) demonstrates CAAPO's superior spatial distribution uniformity and near-complete coverage (98.7% area coverage at $T = 1500$), with only marginal residual gaps. In contrast, APO exhibits persistent coverage voids and nodal redundancy across both iteration thresholds. Benchmark algorithms show analogous limitations in 3D coverage optimization. These empirical results confirm CAAPO's enhanced capability to balance exploration-exploitation trade-offs in complex 3D sensor deployment scenarios through adaptive inertia modulation and chaotic initialization strategies.

6.3. Deployment of 3D sensors

To further validate the usefulness of the CAAPO method in the WSN coverage optimization problem, this part will replicate the

3D WSN deployment process. Assume the solution space is partitioned into a $50 \times 50 \times 50$ cubic area with $N = 30$ nodes, $r = 10$ sensing radius, and 30 population size $P = 30$. To further show CAAPO's capacity to address the coverage problem in WSNs, set the maximum number of iterations to 500 and 1000, respectively, to allow the algorithm to achieve a convergence state.

Comparative convergence analysis (Figure 14) reveals that the baseline APO algorithm exhibits persistent local optima entrapment across both experimental cycles, requiring 200 iterations for partial escape. Conversely, the enhanced CAAPO methodology demonstrates superior local optima avoidance through lens imaging inversion mechanisms, achieving immediate exit upon encountering suboptimal solutions. Quantitative evaluation (Table 7) demonstrates CAAPO's coverage improvement of 10.74% at ($T = 500$) and 0.25% at ($T = 1000$) over APO, with accelerated convergence efficiency attributed to chaotic initialization strategies enhancing population diversity. This methodology reduces computational overhead while improving nodal distribution uniformity (Figure 15), achieving 92.3% coverage efficiency with minimal residual voids.

While benchmark algorithms exhibit significant coverage redundancy and detection gaps, CAAPO maintains an optimal balance between exploration breadth and exploitation depth. These

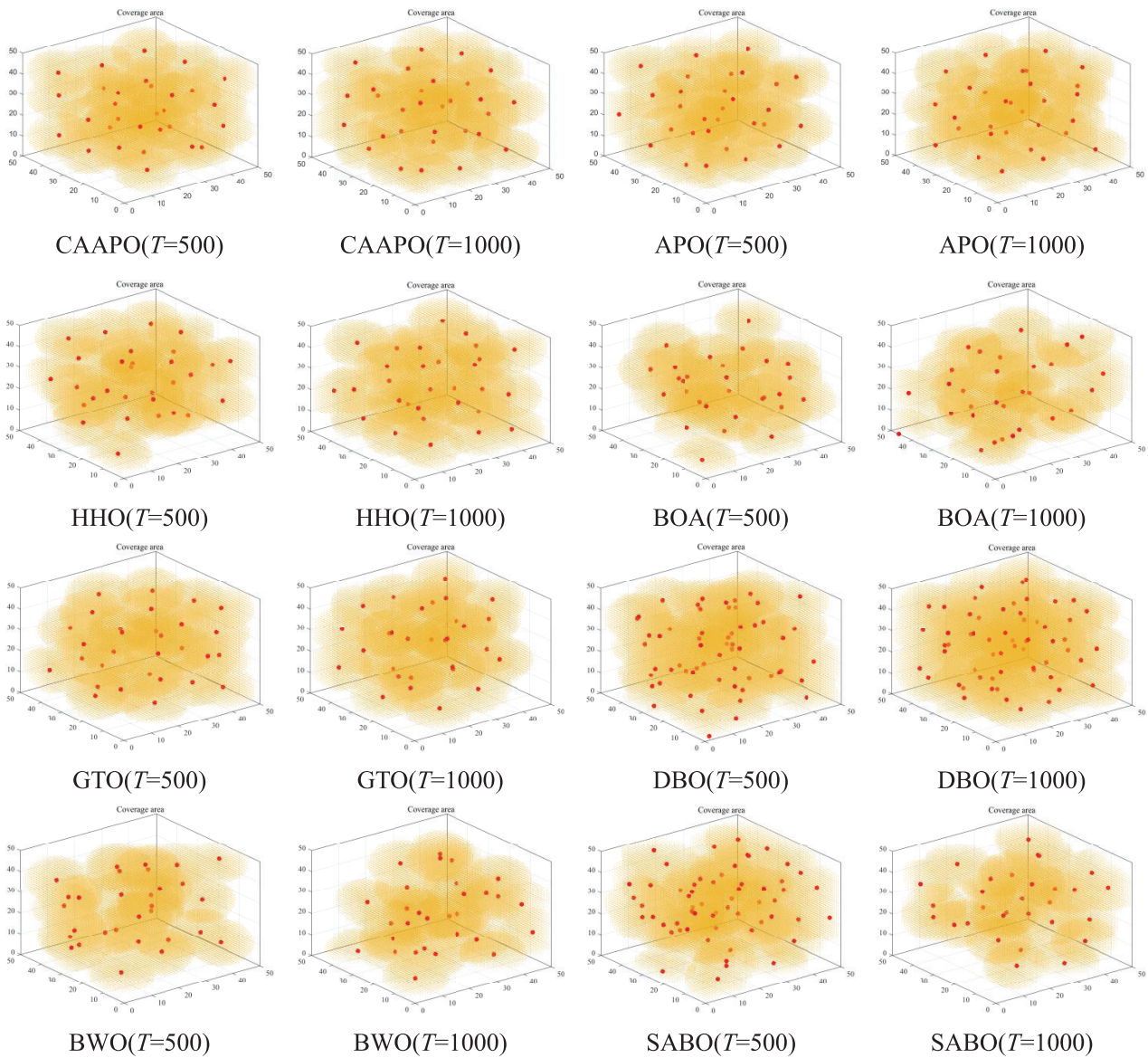


Figure 15: WSN coverage effectiveness diagram.

empirical results validate the technical viability of CAAPO's adaptive framework for 3D WSN deployment optimization.

7. Conclusion and Prospect

In this work, an improved Arctic Puffin Optimization Algorithm (CAAPO) is used to provide a methodological foundation for node placement optimization in multidimensional WSNs. The proposed CAAPO incorporates three strategic modifications to address critical limitations of the baseline APO algorithm: (1) tent chaotic mapping for enhanced initial population quality and accelerated early-phase convergence, (2) lens imaging inversion mechanisms enabling effective local optima circumvention during exploration phases, and (3) adaptive inertia weighting operators that dynamically balance exploration-exploitation trade-offs throughout evolutionary cycles. Empirical validation demonstrates CAAPO's superior performance in maintaining solution diversity while achieving faster convergence rates compared to

conventional approaches, particularly in complex 3D deployment scenarios with heterogeneous terrain constraints.

Empirical validation through benchmark simulations and engineering case studies confirms the enhanced convergence precision of the proposed methodology. The greater effectiveness of CAAPO in both planar and volumetric designs is demonstrated by its subsequent application to sensor network deployment optimization, which maximizes coverage efficiency while reducing resource consumption. These results establish a novel computational paradigm for addressing large-scale high-dimensional optimization challenges in intelligent sensor networks, effectively balancing exploration breadth with exploitation depth across heterogeneous deployment environments.

Addressing current methodological constraints, future research should explore CAAPO's extension to broader optimization domains to enhance cross-domain applicability and practical utility. Concurrently, ongoing investigations will focus on emerging paradigms in intelligent sensor networks, pursuing algorithm-

mic refinements that align with domain-specific innovations to strengthen real-world problem-solving capabilities.

Conflicts of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Author Contributions

Xiangkun Wang: Investigation, Software. **Li Cao:** Formal analysis, Writing – original draft. **Yinggao Yue:** Formal analysis, Methodology. **Zhengwei Zhang:** Writing – original draft. **Tao Zhang:** Writing – review & editing. **Zihui Wang:** Writing – review & editing.

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Data Availability

The data that support the findings of this study are available from the corresponding author upon request. There are no restrictions on data availability.

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Li Cao and Xiangkun Wang contributed equally to this work and should be considered as co-first authors.

References

- Abdollahzadeh, B., Soleimanian Gharehchopogh, F., & Mirjalili, S. (2021). Artificial gorilla troops optimizer: A new nature-inspired metaheuristic algorithm for global optimization problems. *International Journal of Intelligent Systems*, **36**, 5887–5958. <https://doi.org/10.1002/int.22535>.
- Ahmad, I., Adnan, M., ul Amin, N., Khan, T. A., & Shao, J. (2024). Adaptive and priority-based data aggregation and scheduling model for wireless sensor network. *Knowledge-Based Systems*, **303**, 112393. <https://doi.org/10.1016/j.knosys.2024.112393>
- Akraam, M., Rashid, T., & Zafar, S. (2023). An image encryption scheme proposed by modifying chaotic tent map using fuzzy numbers. *Multimedia Tools and Applications*, **82**, 16861–16879. <https://doi.org/10.1007/s11042-022-14053-x>
- Al-Daraiseh, A., Sanjalawe, Y., Al-E'mari, S., Al-Hazaimah, O. M., & Al-Jamal, M. Z. (2023). Cryptographic grade chaotic random number generator based on tent-map. *Journal of Sensor and Actuator Networks*, **12**, 73. <https://doi.org/10.3390/jsan12050073>
- Ammari, H. M. (2021). Connected k-coverage in two-dimensional wireless sensor networks using hexagonal slicing and area stretching. *Journal of Parallel and Distributed Computing*, **153**, 89–109. <https://doi.org/10.1016/j.jpdc.2021.03.011>
- Anka, F. (2025). A novel hybrid metaheuristic method for efficient decentralized IoT network layouts. *Internet of Things*, **32**, 101612. <https://doi.org/10.1016/j.iot.2025.101612>
- Arora, S., & Singh, S. (2019). Butterfly optimization algorithm: A novel approach for global optimization. *Soft Computing*, **23**, 715–734. <https://doi.org/10.1007/s00500-018-3102-4>
- Bahadur, D. K. J., & Lakshmanan, L. (2023). A novel method for optimizing energy consumption in wireless sensor network using genetic algorithm. *Microprocessors and Microsystems*, **96**, 104749. <https://doi.org/10.1016/j.micpro.2023.104749>
- Bai, Y., Cao, L., Chen, B., Chen, Y., & Yue, Y. (2023). A novel topology optimization protocol based on an improved crow search algorithm for the perception layer of the Internet of Things. *Biomimetics*, **8**, 165. <https://doi.org/10.3390/biomimetics8020165>
- Cao, D., Xu, Y., Yang, Z., Dong, Y., & Li, X. (2023). An enhanced whale optimization algorithm with improved dynamic opposite learning and adaptive inertia weight strategy. *Complex & Intelligent Systems*, **9**, 767–795. <https://doi.org/10.1007/s40747-022-00838-y>
- Chen, B., Cao, L., Chen, C., Chen, Y., & Yue, Y. (2024). A comprehensive survey on the chicken swarm optimization algorithm and its applications: State-of-the-art and research challenges. *Artificial Intelligence Review*, **57**, 170. <https://doi.org/10.1007/s10462-024-10770-x>
- Chen, H. W., & Liang, C. K. (2022). Genetic algorithm versus discrete particle swarm optimization algorithm for energy-efficient moving object coverage using mobile sensors. *Applied Sciences*, **12**, 3340. <https://doi.org/10.3390/app12073340>
- Ding, S., Zhang, T., Chen, C., Wang, Y., & Yue, Y. (2023). An efficient particle swarm optimization with evolutionary multi-tasking for stochastic area coverage of heterogeneous sensors. *Information Sciences*, **645**, 119319. <https://doi.org/10.1016/j.ins.2023.119319>
- Du, W., Ying, W., Yang, P., Yao, Y., & Wu, J. (2019). Network-based heterogeneous particle swarm optimization and its application in UAV communication coverage. *IEEE Transactions on Emerging Topics in Computational Intelligence*, **4**, 312–323. <https://doi.org/10.1109/TETCI.2019.2899604>
- Du, Y. (2020). Method for the optimal sensor deployment of WSNs in 3D terrain based on the DPSOVF algorithm. *IEEE Access*, **8**, 140806–140821. <https://doi.org/10.1109/ACCESS.2020.3013106>
- El-Shafeiy, E., Sallam, K. M., Chakraborty, R. K., & Ryan, M. J. (2021). A clustering based swarm intelligence optimization technique for the Internet of Medical Things. *Expert Systems with Applications*, **173**, 114648. <https://doi.org/10.1016/j.eswa.2021.114648>
- Fakhouri, H. N., Alkhalaileh, M. S., Hamad, F., Alawadi, S., Al-Asir, A., Al-Shdifat, A., Al-Husban, B., Alshorman, O., Alshorman, S., & Alshboul, Y. (2024). Hybrid Arctic Puffin Tent Map Algorithm for Solving Design optimization Problems. *Algorithms*, **17**, 589. <https://doi.org/10.3390/a17120589>
- Gad, A. G. (2022). Particle swarm optimization algorithm and its applications: A systematic review. *Archives of Computational Methods in Engineering*, **29**, 2531–2561. <https://doi.org/10.1007/s11831-021-09675-7>
- Gao, X., Zhu, X., & Zhai, L. (2023). AoI-sensitive data collection in multi-UAV-assisted wireless sensor networks. *IEEE Transactions on Wireless Communications*, **22**, 5185–5197. <https://doi.org/10.1109/TWC.2023.3237686>
- Gezici, H., & Livatyali, H. (2022). Chaotic Harris hawks optimization algorithm. *Journal of Computational Design and Engineering*, **9**, 216–245. <https://doi.org/10.1093/jcde/qwab082>
- Gou, P., Guo, B., Guo, M., Wang, W., & Zhou, T. (2023). VKECE-3D: Energy-efficient coverage enhancement in three-dimensional heterogeneous wireless sensor networks based on 3D-Voronoi and K-means algorithm. *Sensors*, **23**, 573. <https://doi.org/10.3390/s23020573>

- Hanh, N. T., Binh, H. T. T., Hoai, N. X., & Palaniswami, M. S. (2019). An efficient genetic algorithm for maximizing area coverage in wireless sensor networks. *Information Sciences*, **488**, 58–75. <https://doi.org/10.1016/j.ins.2019.02.056>
- Heidari, A. A., Mirjalili, S., Faris, H., Aljarah, I., Mafarja, M., & Chen, H. (2019). Harris hawks optimization: Algorithm and applications. *Future Generation Computer Systems*, **97**, 849–872. <https://doi.org/10.1016/j.future.2019.02.028>
- Jaafari, A., Panahi, M., Mafi-Gholami, D., Gholamnia, F., & Lee, S. (2022). Swarm intelligence optimization of the group method of data handling using the cuckoo search and whale optimization algorithms to model and predict landslides. *Applied Soft Computing*, **116**, 108254. <https://doi.org/10.1016/j.asoc.2021.108254>
- Jena, J. J., & Satapathy, S. C. (2024). A new adaptive tuned Social Group optimization (SGO) algorithm with sigmoid-adaptive inertia weight for solving engineering design problems. *Multimedia Tools and Applications*, **83**, 3021–3055. <https://doi.org/10.1007/s11042-023-15683-5>
- Jia, H., Li, Y., Wu, D., Wang, S., & Jiang, S. (2023). Multi-strategy remora optimization algorithm for solving multi-extremum problems. *Journal of Computational Design and Engineering*, **10**, 1315–1349. <https://doi.org/10.1093/jcde/qwad053>
- Jiang, S., Yue, Y., Chen, C., Chen, Y., & Cao, L. (2024). A multi-objective optimization problem solving method based on improved golden jackal optimization algorithm and its application. *Biomimetics*, **9**, 270. <https://doi.org/10.3390/biomimetics9050270>
- Kanwal, S., Inam, S., Othman, M. T. B., Alshammari, F. S., & Al-Harbi, T. S. (2022). An effective color image encryption based on Henon map, tent chaotic map, and orthogonal matrices. *Sensors*, **22**, 4359. <https://doi.org/10.3390/s22124359>
- Kiani, F. (2025). A multi-objective metaheuristic method for node placement in dynamic IoT environments. *Discover Internet of Things*, **5**, 60. <https://doi.org/10.1007/s43926-025-00060-5>
- Kiani, F., & Seyyedabbasi, A. (2022). Metaheuristic algorithms in IoT: Optimized edge node localization. In: Seyyedabbasi A., & Kiani F. (Eds.), *Engineering applications of modern metaheuristics* (pp. 19–39). Springer International Publishing. https://doi.org/10.1007/978-3-030-93616-4_2
- Kivi, M. E., & Majidnezhad, V. (2022). A novel swarm intelligence algorithm inspired by the grazing of sheep. *Journal of Ambient Intelligence and Humanized Computing*, **13**, 1201–1213. <https://doi.org/10.1007/s12652-021-03661-4>
- Li, G., Zhang, T., Tsai, C. Y., Pan, J. S., & Chu, S. C. (2024). Modified crayfish optimization algorithm with adaptive spiral elite greedy opposition-based learning and search-hide strategy for global optimization. *Journal of Computational Design and Engineering*, **11**, 249–305. <https://doi.org/10.1093/jcde/qwae030>
- Li, J., Gao, M., Pan, J. S., & Chu, S. C. (2021). A parallel compact cat swarm optimization and its application in DV-Hop node localization for wireless sensor network. *Wireless Networks*, **27**, 2081–2101. <https://doi.org/10.1007/s11276-021-02554-w>
- Li, J., Li, G. C., Chu, S. C., & Pan, J. S. (2022). Modified parallel tunicate swarm algorithm and application in 3D WSNs coverage optimization. *Journal of Internet Technology*, **23**, 227–244. <https://doi.org/10.53106/160792642022032302001>
- Li, Q., & Liu, N. (2020). Monitoring area coverage optimization algorithm based on nodes perceptual mathematical model in wireless sensor networks. *Computer Communications*, **155**, 227–234. <https://doi.org/10.1016/j.comcom.2020.03.006>
- Nematzadeh, S., Torkamanian-Afshar, M., Seyyedabbasi, A., & Kiani, F. (2023). Maximizing coverage and maintaining connectivity in WSN and decentralized IoT: An efficient metaheuristic-based method for environment-aware node deployment. *Neural Computing and Applications*, **35**, 611–641. <https://doi.org/10.1007/s00521-022-07794-1>
- Ouyang, C., Zhu, D., & Qiu, Y. (2021). Lens learning sparrow search algorithm. *Mathematical Problems in Engineering*, **2021**, 9935090. <https://doi.org/10.1155/2021/9935090>
- Priyadarshi, R., & Gupta, B. (2021). Area coverage optimization in three-dimensional wireless sensor network. *Wireless Personal Communications*, **117**, 843–865. <https://doi.org/10.1007/s11277-020-07898-8>
- Rahaman, M. M., & Azharuddin, M. (2022). Wireless sensor networks in agriculture through machine learning: A survey. *Computers and Electronics in Agriculture*, **197**, 106928. <https://doi.org/10.1016/j.compag.2022.106928>
- Rani, K. P., Sreedevi, P., Poornima, E., Pradeep, G. N., & Kumar, N. S. (2023). FTOR-Mod PSO: A fault tolerance and an optimal relay node selection algorithm for wireless sensor networks using modified PSO. *Knowledge-Based Systems*, **272**, 110583. <https://doi.org/10.1016/j.knosys.2023.110583>
- Siamantas, G., & Kandris, D. (2024). Particle swarm optimization for k-coverage and 1-connectivity in wireless sensor networks. *Electronics*, **13**, 4841. <https://doi.org/10.3390/electronics13234841>
- Singh, P., Mittal, N., & Singh, P. (2022). A novel hybrid range-free approach to locate sensor nodes in 3D WSN using GWO-FA algorithm. *Telecommunication Systems*, **80**, 303–323. <https://doi.org/10.1007/s11235-022-00920-3>
- Temel, S., Unaldi, N., & Kaynak, O. (2013). On deployment of wireless sensors on 3-D terrains to maximize sensing coverage by utilizing cat swarm optimization with wavelet transform. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, **44**, 111–120. <https://doi.org/10.1109/TSMCC.2013.2258333>
- Temene, N., Sergiou, C., Georgiou, C., & Vassiliou, V. (2022). A survey on mobility in wireless sensor networks. *Ad Hoc Networks*, **125**, 102726. <https://doi.org/10.1016/j.adhoc.2021.102726>
- Tossa, F., Abdou, W., Ansari, K., & Raimy, X. (2022). Area coverage maximization under connectivity constraint in wireless sensor networks. *Sensors*, **22**, 1712. <https://doi.org/10.3390/s22051712>
- Trojovský, P., & Dehghani, M. (2023). Subtraction-average-based optimizer: A new swarm-inspired metaheuristic algorithm for solving optimization problems. *Biomimetics*, **8**, 149. <https://doi.org/10.3390/biomimetics8020149>
- Walia, G. S., Singh, P., Singh, M., & Goyal, A. (2022). Three dimensional optimum node localization in dynamic wireless sensor networks. *CMC-Computers, Materials & Continua*, **70**, 305–321. <https://doi.org/10.32604/cmc.2022.019324>
- Wang, W., Tian, W., Xu, D., Li, G., & Zhang, L. (2024). Arctic puffin optimization: A bio-inspired metaheuristic algorithm for solving engineering design optimization. *Advances in Engineering Software*, **195**, 103694. <https://doi.org/10.1016/j.advengsoft.2024.103694>
- Wang, Y., Wu, S., Chen, Z., Gao, X., & Chen, G. (2017). Coverage problem with uncertain properties in wireless sensor networks: A survey. *Computer Networks*, **123**, 200–232. <https://doi.org/10.1016/j.comnet.2017.05.011>
- Xue, J., & Shen, B. (2023). Dung beetle optimizer: A new metaheuristic algorithm for global optimization. *The Journal of Supercomputing*, **79**, 7305–7336. <https://doi.org/10.1007/s11227-022-04959-6>
- Yang, T., Jiang, X., Huang, Y., & Yang, J. (2022). Mechanical sensors based on two-dimensional materials: Sensing mechanisms, structural designs and wearable applications. *Iscience*, **25**, 103728. <https://doi.org/10.1016/j.isci.2021.103728>
- Yao, Y., Liao, H., Liu, M., Wang, J., & Yue, Y. (2023). Coverage optimization strategy for 3-D wireless sensor networks based on improved

- sparrow search algorithm. *IEEE Sensors Journal*, **23**, 23721–23733. <https://doi.org/10.1109/JSEN.2023.3305010>
- Yarinezhad, R., & Hashemi, S. N. (2023). A sensor deployment approach for target coverage problem in wireless sensor networks. *Journal of Ambient Intelligence and Humanized Computing*, **14**, 5941–5956. <https://doi.org/10.1007/s12652-020-02657-w>
- Yue, Y., Cao, L., Lu, D., Huang, S., & Wang, S. (2023). Review and empirical analysis of sparrow search algorithm. *Artificial Intelligence Review*, **56**, 10867–10919. <https://doi.org/10.1007/s10462-023-10435-1>
- Yue, Y., Cao, L., & Zhang, Y. (2024). Novel WSN coverage optimization strategy via monarch butterfly algorithm and particle swarm optimization. *Wireless Personal Communications*, **135**, 2255–2280. <https://doi.org/10.1007/s11277-024-11350-6>
- Zhang, L., & Chen, X. B. (2024). Enhanced chimp hierarchy optimization algorithm with adaptive lens imaging for feature selection in data classification. *Scientific Reports*, **14**, 6910. <https://doi.org/10.1038/s41598-024-57659-x>
- Zhang, L. G., Fan, F., Chu, S. C., & Pan, J. S. (2021). Hybrid strategy of multiple optimization algorithms applied to 3-D terrain node coverage of wireless sensor network. *Wireless Communications and Mobile Computing*, **2021**, 6690824. <https://doi.org/10.1155/2021/6690824>
- Zhang, Y. (2020). Coverage optimization and simulation of wireless sensor networks based on particle swarm optimization. *International Journal of Wireless Information Networks*, **27**, 307–316. <https://doi.org/10.1007/s10776-020-00476-6>
- Zheng, R., Hussien, A. G., Qaddoura, R., & Alomari, O. A. (2023). A multi-strategy enhanced African vultures optimization algorithm for global optimization problems. *Journal of Computational Design and Engineering*, **10**, 329–356. <https://doi.org/10.1093/jcde/qwac135>
- Zhong, C., Li, G., & Meng, Z. (2022). Beluga whale optimization: A novel nature-inspired metaheuristic algorithm. *Knowledge-Based Systems*, **251**, 109215. <https://doi.org/10.1016/j.knsys.2022.109215>